Reaction types in Organometallic Chemistry

—Elementary Reactions in Organometallic Chemistry (基元反应)

(1) Ligand coordination and substitution √

(配体的配位和取代)

(2) Oxidative addition and reductive elimination

(氧化加成和还原消除)

(3) Insertion and elimination (de-insertion) reactions

(插入反应和消除(反插入)反应)

(4) Electrophilic and Nucleophilic Addition and Abstraction

(亲电和亲核加成与攫取)

(Reactions on the complexed ligands 和金属结合的配体的反应)

Help us to understand and predict the metal-catalyzed or mediated reactions.

Be aware:

- Classified **in terms of reaction patterns**, not mechanism !!
- An elementary reaction may go through different mechanisms.

Chapter 6. Oxidative Addition and Reductive Elimination

Outline

A. Introduction to oxidative addition reactions

B. Mechanism of oxidative addition reactions

C. Examples of addition of specific molecules

D. Reductive elimination

E. Oxidative coupling and reductive fragmentation

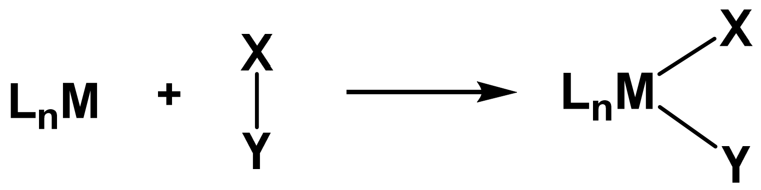
References and suggested readings:

1. *The Organometallic Chemistry of the Transition Metals*, Robert H. Crabtree, 6th Edition, **2014**, Chapter 6.
2. *《过渡金属有机化学》*江焕峰, 祝诗发 译, **2012**, 科学出版社.
3. *Organometallic Chemistry*, G. O. Spessard, G. L. G. L. Miessler, Prentice-Hall: New Jersey, **1997**, Chapter 7.
4. *Organotransition Metal Chemistry*, Akio Yamamoto, **1986**, Chapter 6.2.

A. Introduction to oxidative addition reactions

Q1. What is an oxidative addition reaction?

Most common cases

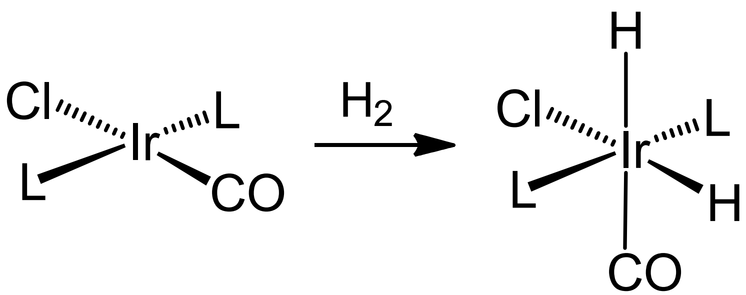


16e → 18e

ΔO.S. = 2

C.N. = n → C.N. = n + 2

Example:



However, there are a few other types reactions that can be also regarded as oxidative addition reactions.

A few other types reactions that can be also regarded as oxidative addition reactions.

***a. Electrophilic attack of metals.* e.g.**

$$L_nM + E^+ \longrightarrow L_n\overset{+}{M}{-}E$$

ΔO.S. = 2

ΔC.N. = 1

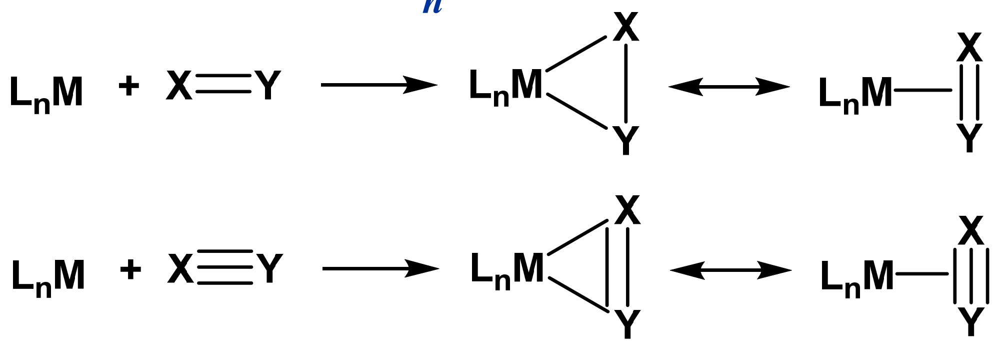
No change in total valence electrons count !

Examples: $Fe(CO)_5 + H^+ \longrightarrow [HFe(CO)_5]^+$

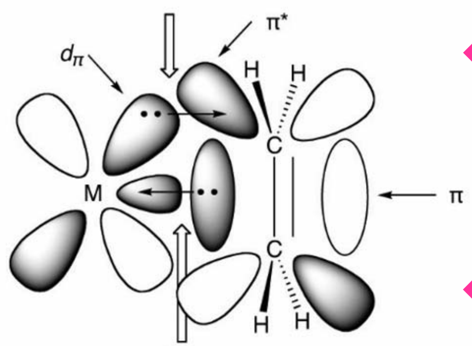
$[Mn(CO)_5]^- + CH_3I \longrightarrow CH_3Mn(CO)_5 + I^-$

A few other types reactions that can be also regarded as oxidative addition reactions.

b. Addition of X=Y or X≡Y to L_nM.

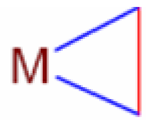


Recall bonding in alkene complexes



π-backbonding: e- from dπ(M) ---> π*(olefin)

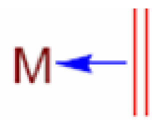
- **e^- rich systems: π-backdonation is dominant.**



X_2-Type
metallacyclopropane

Metal is oxidized formally

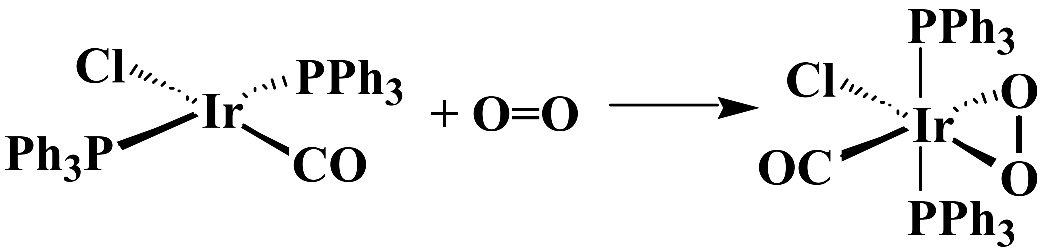
- **e- poor systems: σ-donation is dominant.**



L-Type
π-complex

σ donation: e- from π(olefin) --->σ (M)

Additional example:



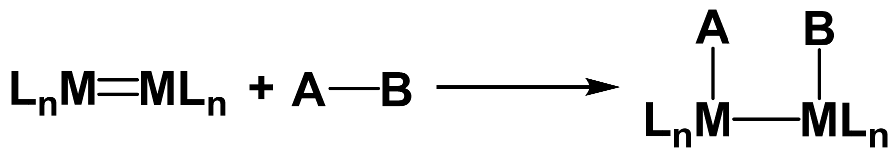
A few other types reactions that can be also regarded as oxidative addition reactions.

c. Binuclear oxidation reactions.

$$\underset{18e}{L_nM{-}ML_n} + A{-}B \longrightarrow \underset{18e}{L_nM{-}A} + \underset{18e}{L_nM{-}B}$$

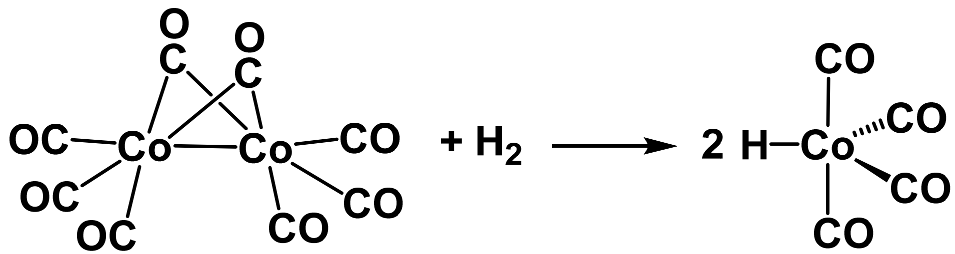
ΔO.S. = 1

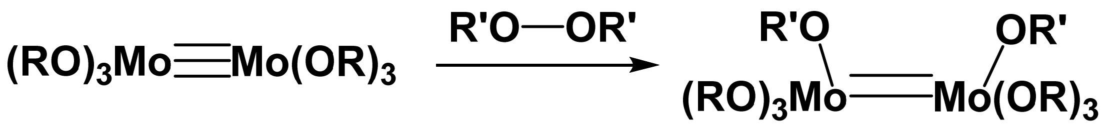
Note: no change in valence electron count



ΔO.S. = 1

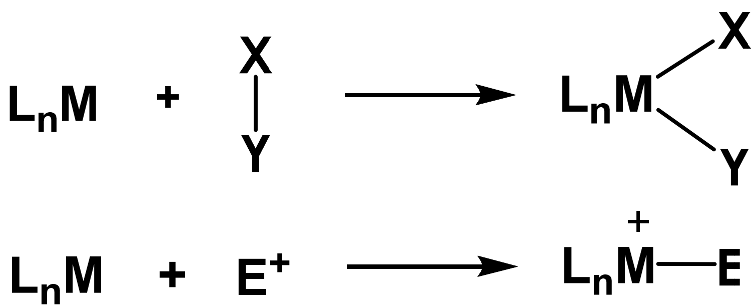
Examples:





A. Introduction to oxidative addition reactions

Q2. What types of complexes L_nM can undergo oxidative addition reactions?



*** Higher oxidation state is accessible stable.**

*** Availability of vacant site for newly formed bonds.**

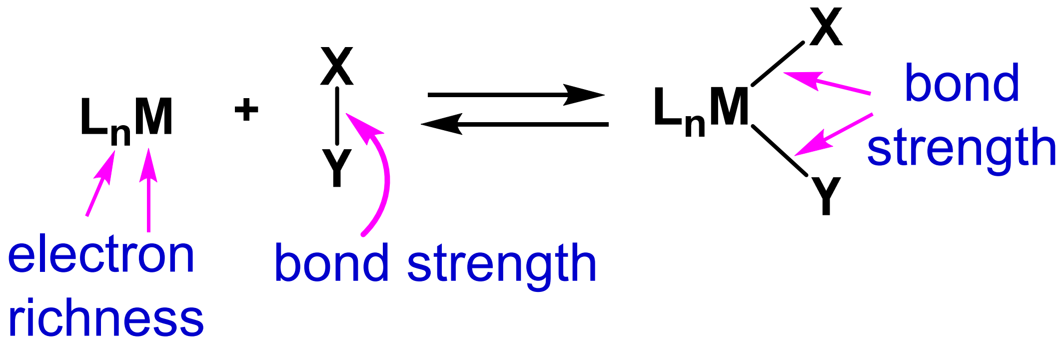
Can the following complexes undergo O.A. reactions?

TiR_4 WR_6

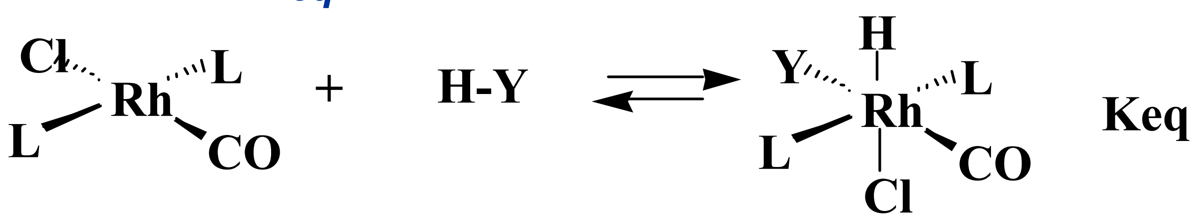
No, Ti(IV) W(VI)

A. Introduction to oxidative addition reactions

Q3. In general, what are the factors affect the following equilibrium?



Q3A. Rank the order of K_{eq} for the reactions below.

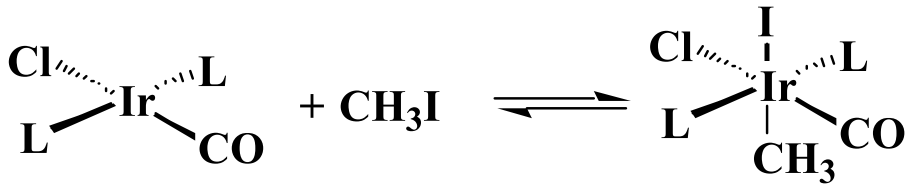


L = PPh_3, PMe_3, $P(OPh)_3$

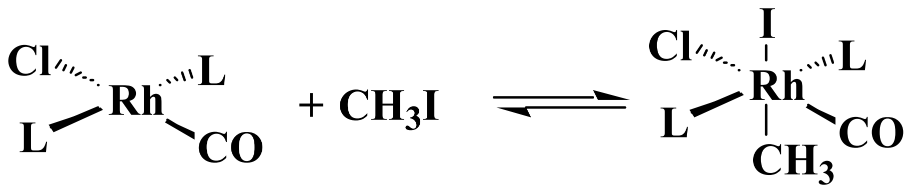
Order in Keq: $P(OPh)_3 < PPh_3 < PMe_3$

Electron-rich metal center favor oxidative addition product.

Q3B. Which of the following reactions has a larger Keq?



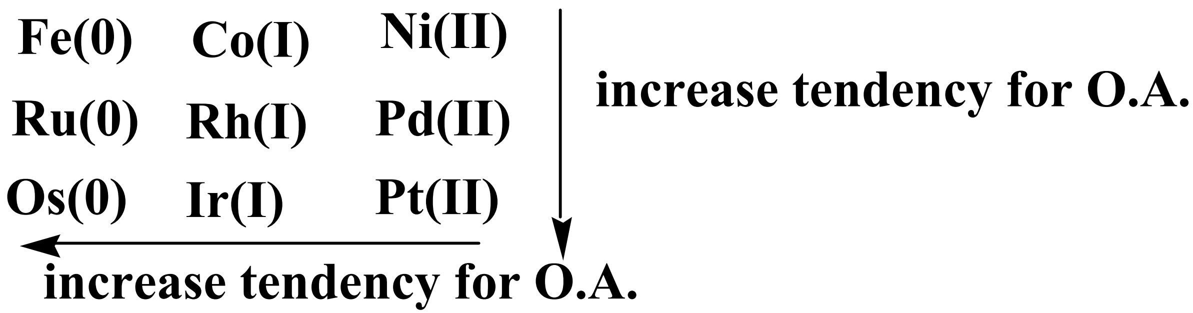
(a)



(b)

Answer: (a). The energy of d electron of Ir is relatively higher.

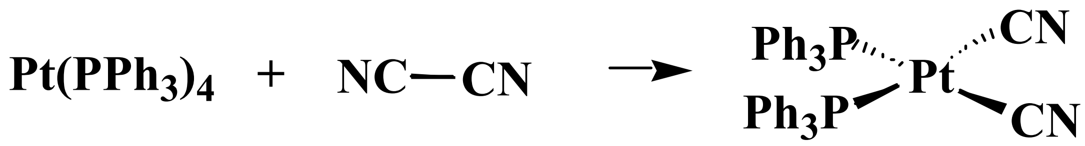
In fact, for isostructural d^8 complexes:



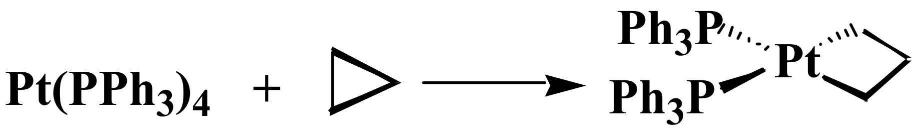
This is the trend for d orbital energy and ability for back-donation.

Q3C. Which of the following reaction is least likely?

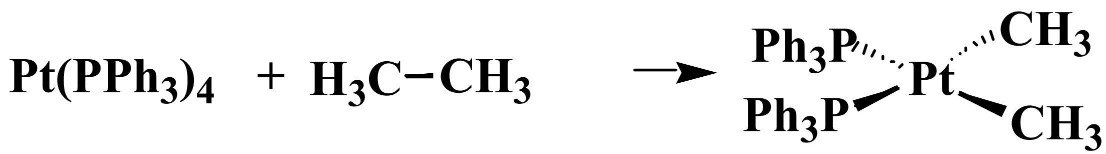
(a)



(b)



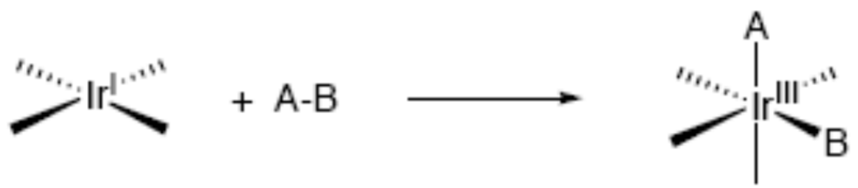
(c)



Answer: (c). Pt-CH_3 bond is weaker, H_3C-CH_3 bond is strong.

Weak X-Y bond favors O.A. reaction.
Stable M-X, M-Y bonds favor O.A. reaction.

Thermochemistry of Oxidative Addition

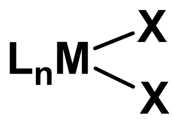


A-B	A-B BDE (kcal/mol)	Ir-A BDE (kcal/mol)	Ir-B BDE (kcal/mol)	ΔH (kcal/mol)	ΔG (kcal/mol)
H-H	104	Ir-H: 60	Ir-H: 60	-16	-6
CH_3-H	104	Ir-CH_3: 46	Ir-H: 60	-2	+8
CH_3-CH_3	88	Ir-CH_3: 46	Ir-CH_3: 46	-4	+6
CH_3-I	56	Ir-CH_3: 46	Ir-I: 45	-35	-25

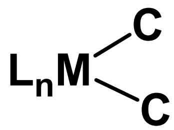
A. Introduction to oxidative addition reactions

Q4. What bonds can be oxidatively added?

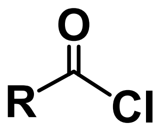
X-X H_2, Cl_2, Br_2, I_2, RS-SR, RO-OR

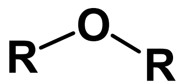


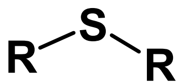
C-C Ph_3C-C-Ph_3, NC-CN, Ph-CN

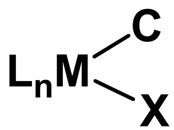


C-X CH_3I, Ph-I, CH_2Cl_2, CCl_4

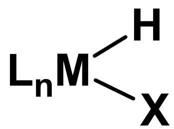




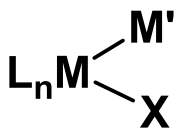




H-X HCl, HBr, HI, RO-H, RS-H,
R_2N-H, R_2P-H, R_2B-H

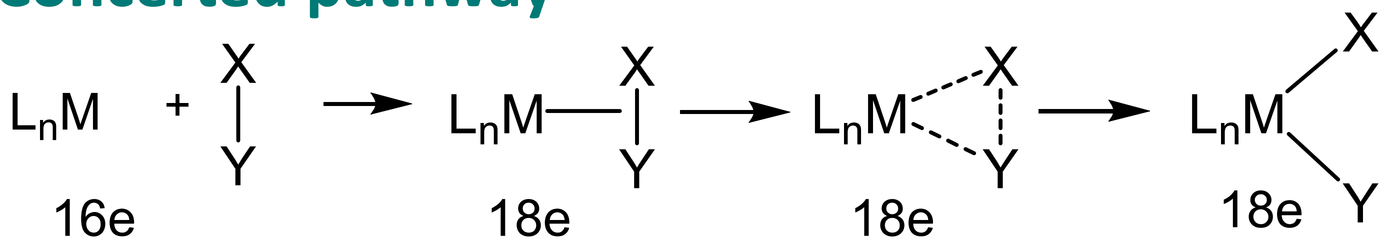


M'-X Ph_3PAu-Cl, Cl-Hg-Cl, R_3Sn-Cl
R_3Si-Cl, Ph_2B-X

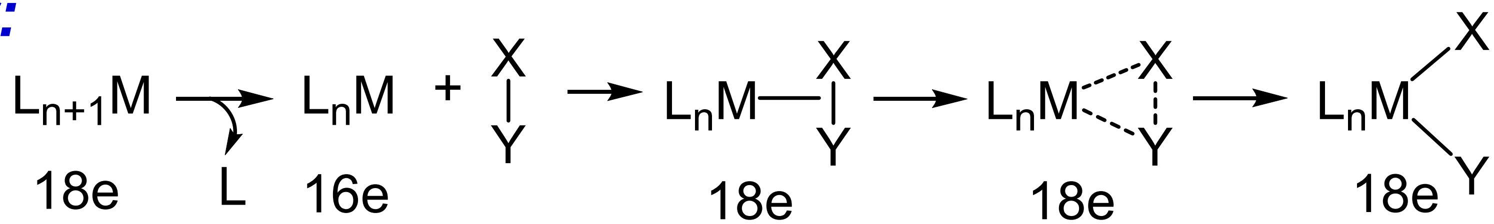


B. Mechanisms of oxidative addition reactions

1). Concerted pathway



or:



Q1. What are the most important features?

- # of valence e- count of reactant L_nM ?   ≤ 16
- $\Delta S^{\#} > 0$ or < 0? < 0
- Any solvent effect on reaction rate?   no

Q2. What is the character of substrate X-Y bonds?

X-Y ---> non-polar or low polar

e. g.   H-H, R_3Si-H, R_3C-H, C-C; H-X in nonpolar solvents.

Mechanisms of oxidative addition reactions

polar reagents.

Table 6.1. Examples of reagents with low polarity, high polarity, and intermediate polarity that undergo oxidative addition to transition metal complexes.

Reagents that are nonpolar or have low polarity:

H_2, RH, ArH, R_3SiH, R_3SnH, $R_3Sn{-}SnR_3$, $R_2B{-}H$, R_2BBR_2, RSSR, NC–CN, and $Ph{-}PPh_2$

Reagents that are highly polar:

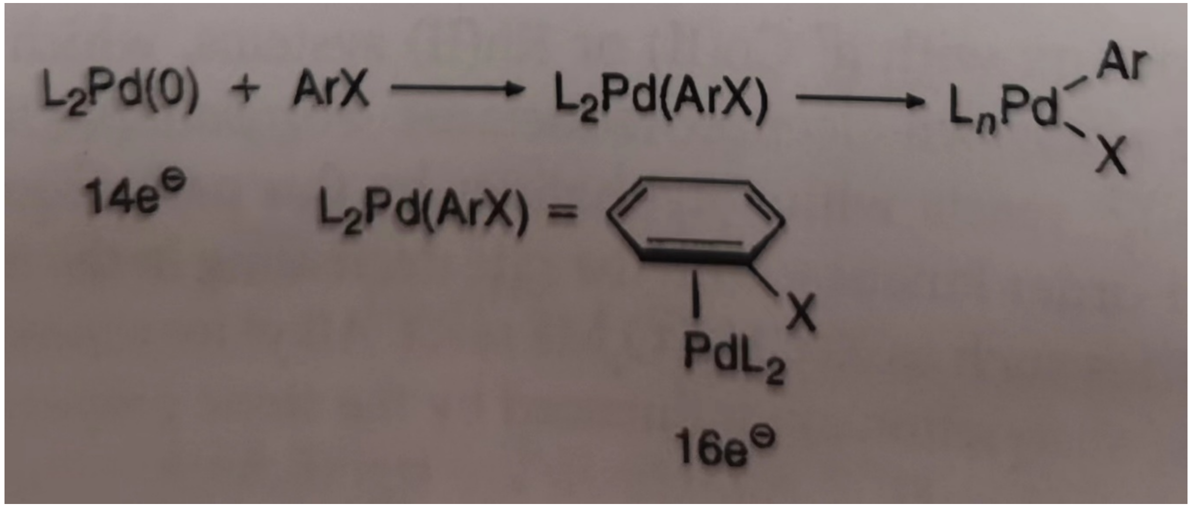
HX, X_2, RCO_2H, RX, ROTs, RC(O)X, RSO_2X, $ROSO_2X$, RC(O)–OPh, HgX_2, and $SnCl_4$, $GeCl_4$, R_3SnCl, R_3PbCl

Reagents with medium polarity:

RSH, ROH, RNH_2, ArX, ArCN, CCl_4, and $CHCl_3$

Concerted Oxidative Additions of Reagents with C-X Bonds of Medium Polarity

Oxidative addition of substrates possessing C-X bonds with medium polarity such as Ar-X bonds often occur by concerted pathways involving three-centered transition states more like those of the oxidative additions of nonpolar substrates.



Aryl halides can undergo oxidative addition via a concerted mechanism.
e.g. $[Pd(P\{Ar\}_3)_3]$ reacts with ArBr in this way. Prior loss of PAr_3 is required to give the very reactive 1-coordinate intermediate, $Pd(PAr_3)$, that goes on to give $[(PAr_3)(Ar))Pd(\mu\text{-}Br)]_2$ as final product.
The more reactive ArI compounds do not need prior dissociation of L and can give OA with $L_2Pd(0)$

Practice Problems

Which of the following oxidative addition reactions is/are likely to proceed through a concerted mechanism?

√ (a) $IrCl(PPh_3)_3 + H_2$ -----------> $IrH_2Cl(PPh_3)_3$

√ (b) $IrCl(PPh_3)_3 + SiH_4$ -----------> $IrH(SiH_3)Cl(PPh_3)_3$

(c) $Pd(PMe_3)_4 + PhI$-----------> $PdI(Ph)(PMe_3)_2$

(cannot generate PdL_2 intermediate)

(d) $Pt(PPh_3)_4 + CH_3I$-----------> $PdI(CH_3)(PPh_3)_2$

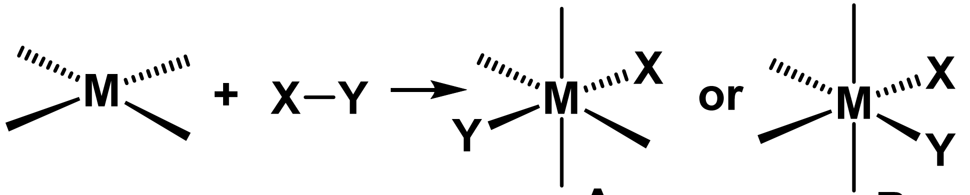
(highly polar)

(e) two of them (f) three of them (g) all of them

(h) none of them

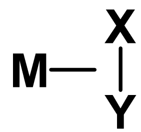
Answer: (e)

Q3. Stereochemistry at M. If via concerted mechanism, which one should be the product?

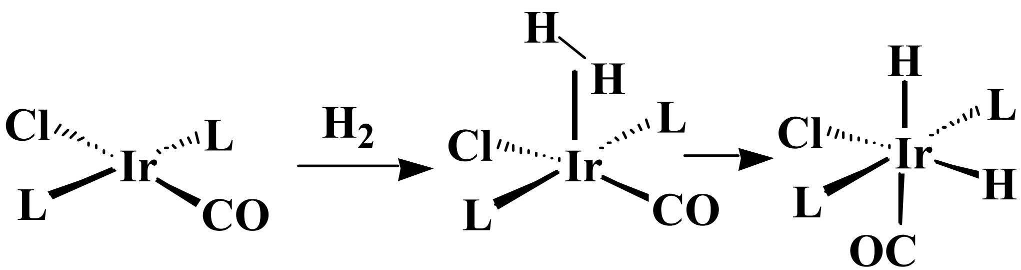


A B

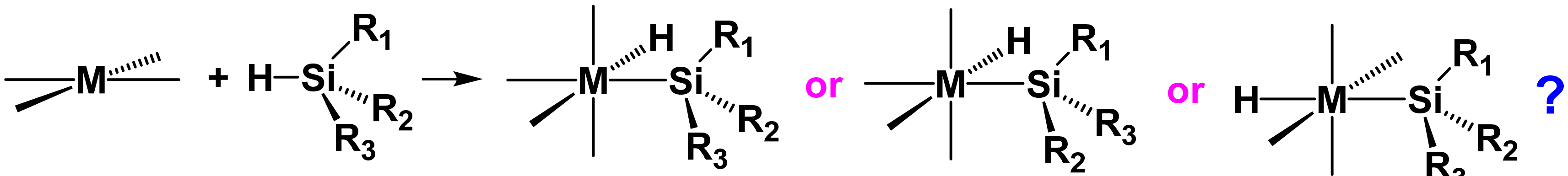
Answer: B. X, Y have to be cis to each other



Example.

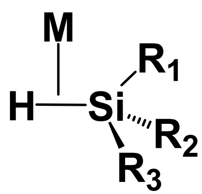


Q4. Stereochemistry at X or Y. If via concerted mechanism, which one should be the product?



A B C

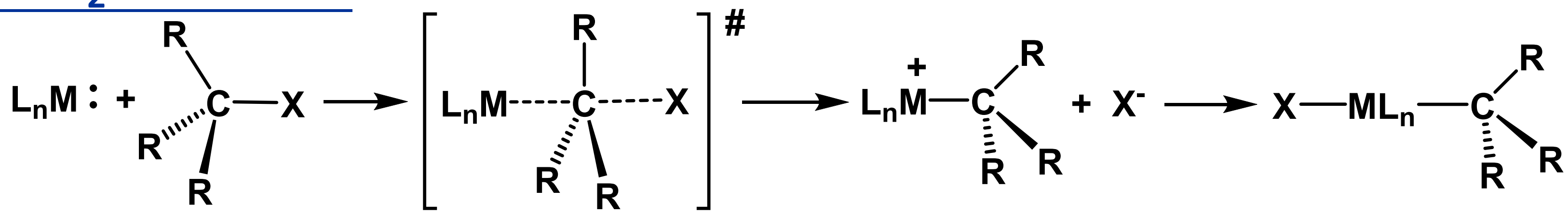
Answer: A. Sterochemistry of Si should be retained, H, Si should be cis to each other.



B. Mechanisms of oxidative addition reactions

2). Non-concerted mechanism

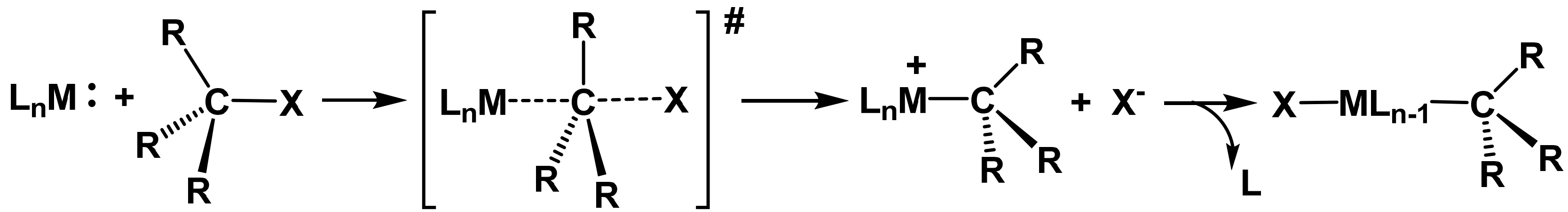
a). SN_2 reactions



16e or less complexes

16e

18e



18e complexes

18e

18e

Q1. **What are the most important features?**

*# of valence e^- count of reactant L_nM — 18 or less

* $\Delta S^{\#} > 0$ or < 0? — < 0

*Any solvent effect on reaction rate? — yes

e.g ***in benzene, acetone same reaction rate?***

Charge separation in TS, reaction is faster in a polar solvent

Q2. What kinds of substrates may undergo O.A. *via* SN_2 Mechanism?

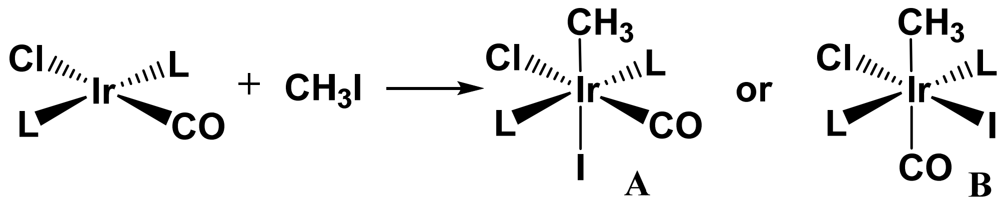
Good electrophiles that can produce stable anions. e.g.

e.g. X_2: Cl_2, Br_2, I_2

R-X: Ar-X, CH_3X, $PhCH_2X$, CH_2=CH-CH_2X

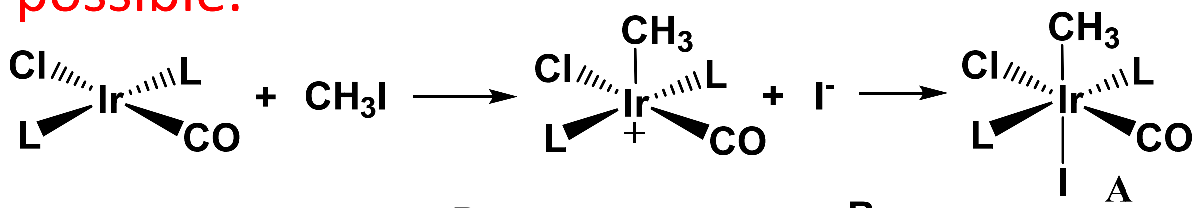
Q3. Stereochemistry at M.

If *via* SN_2 mechanism, which product should one get?

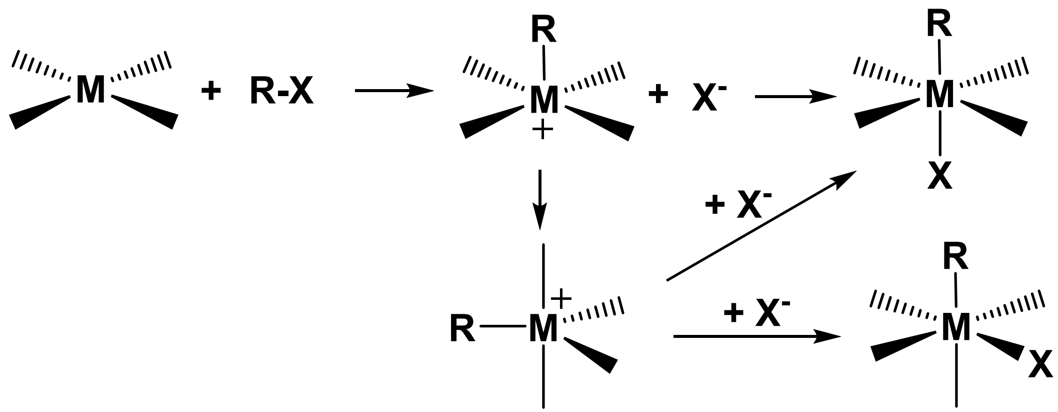


In general, both are possible.

Experimentally **A** was obtained.

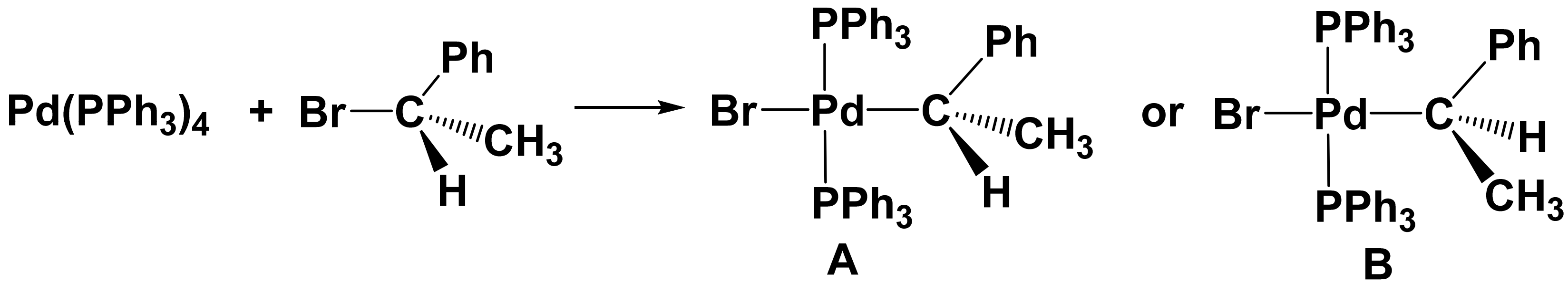


General cases:



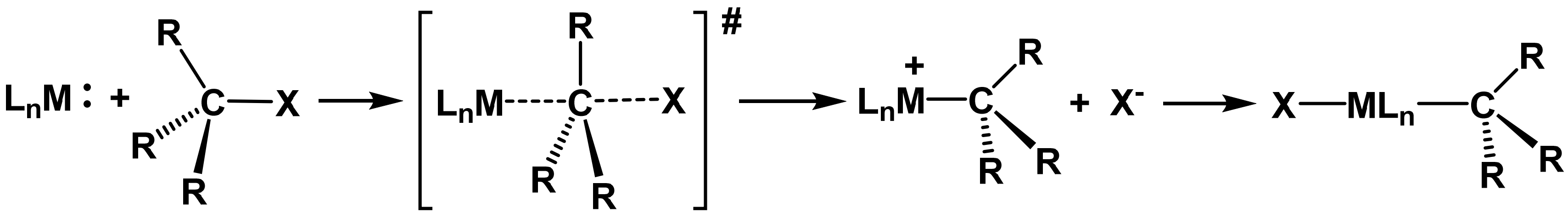
Q4. Stereochemistry at R.

If *via* SN_2 mechanism, what would be the product?



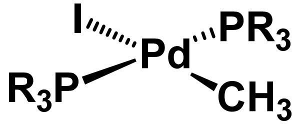
Answer: B. The configuration at C should be inverted.

General cases:



Q5. Relative reaction rate.

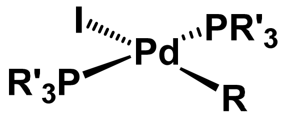
For the pair of oxidative addition reactions below, select the one with **highest reaction rate**. Assume that the reactions are though S_N2 mechanism.

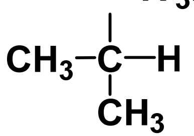
A). $Pd(PR_3)_4 + CH_3I \longrightarrow$ 

a. $PR_3 = PPh_3$

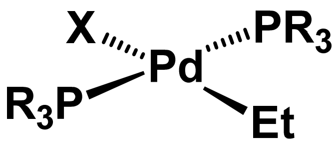
b. $PR_3 = PMe_3$

b. The complex is more e- rich and therefore more nucleophilic.

B). $Pd(PR'_3)_4 + R\text{-}I \longrightarrow$ 

R = CH_3-, CH_3CH_2-, 

R = CH_3. CH_3 has least steric hindrance. Therefore CH_3I is more reactive.

C). $Pd(PR_3)_4 + Et\text{-}X \longrightarrow$ 

X = Cl, I, OTs

X = OTs. OTs- is a better leaving group.

In general, the order of reactivity of R-X are:

R-OTs > R-I > R-Br > R-Cl

B. Mechanisms of oxidative addition reactions

2). Non-concerted mechanism

b). Other non-concerted process

(i) Radical mechanisms

- ***Non-chain radical mechanism***
- ***Chain radical mechanism***

ii) Ionic mechanism

Some molecules (e.g. HCl, HBr, HI) dissociate into ions in polar solvents.

$$HCl + H_2O \text{ ----------> } H^+(aqua) + Cl^- (aqua)$$

Therefore oxidative addition of these molecules in polar solvents must involve ionic species.

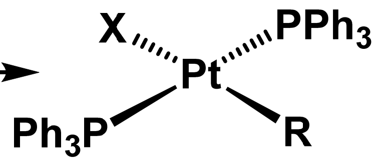
b). Other non-concerted process

(i) Radical mechanisms

- ***Less desirable are oxidative additions involving radicals, because these reactive intermediates tend to give undesired side-reactions.***
- ***Minor changes in the structure of the substrate, the complex, or even the impurity level can be enough to affect the rate.***
- ***The alkyl group always loses any stereochemistry at the α-carbon because the radical is planar.***
- ***In radical reactions, the solvent must not react fast with R· intermediates; alkane, C_6H_6, AcOH, CH_3CN, and water are usually suitable.***

(i) Radical mechanisms

Non-chain radical mechanism, e.g.

$Pt(PPh_3)_3$ + R-X $\longrightarrow$ 

RX = MeI, EtI, $PhCH_2I$, $PhCH_2Br$

(Lappert and Lednor, *J. Chem. Soc., Dalton Trans,* 1980, 1448; *Adv. Organomet. Chem.*, 1976, **14**, 345.)

$$Pt(PPh_3)_3 \underset{+\,PPh_3}{\overset{-\,PPh_3}{\rightleftharpoons}} Pt(PPh_3)_2 \xrightarrow{RX} (Ph_3P)_2Pt\text{---}XR \xrightarrow{slow} {\bullet}PtX(PPh_3)_2 + R{\bullet} \xrightarrow{fast} R\text{-}PtX(PPh_3)_2$$

1e transfers from M to the RX σ to form ·PtX(PPh$_3$)$_2$ and R· radicals*

The radical pair rapidly recombines

- ***Like the SN$_2$ process, the reaction is faster the more basic the metal, and the more readily electron transfer takes place, which gives the reactivity order: RI > RBr > RCl;***

  Unlike the SN$_2$ process, the reaction is very slow for alkyl tosylates [e.g., ROSO$_2$(C$_6$H$_4$Me)] (RCl >> ROTs).

- ***The reaction goes faster as R· becomes more stable and easier to form, giving rise to the reactivity order: Me < 1° < 2° < 3°***

<u>**Survivable Ni(I)XL$_3$ intermediate:**</u> sufficiently stable to survive as an observable reaction product

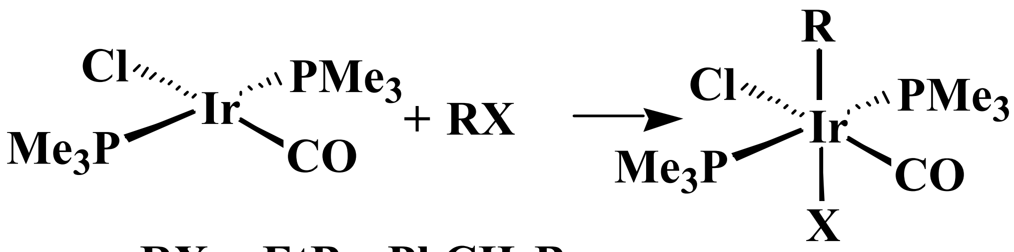
$$NiL_3 + Ar\text{-}X \xrightarrow{solvent} Ni(I)XL_3 + ArH$$

Ar· radical abstracts an H atom from the solvent before it can combine with the $L_2XNi(I)\cdot$ radical.

(i) Radical mechanisms

◆ ***Chain radical mechanism, e.g.***

Vaska's
PMe_3 analog



$RX = EtBr, PhCH_2Br$

$R\text{-}X \text{ ----------------> } R^{\bullet}$

$R^{\bullet} + IrCl(CO)(PMe_3)_2 \text{ ------> } R\overset{\bullet}{I}rCl(CO)(PMe_3)_2$

$R\overset{\bullet}{I}rCl(CO)(PMe_3)_2 + R\text{-}X \text{ ----> } RIrXCl(CO)(PMe_3)_2 + R^{\bullet}$

$R^{\bullet} + R^{\bullet} \text{ ----------> } R\text{-}R$

- ***A radical initiator, Q· (e.g., a trace of air or peroxide in the solvent), may be required to substitute for R· in the first cycle to set the process going.***
- ***Chain termination steps, such as recombination of two R· to give R_2, limits the number of possible cycles.***

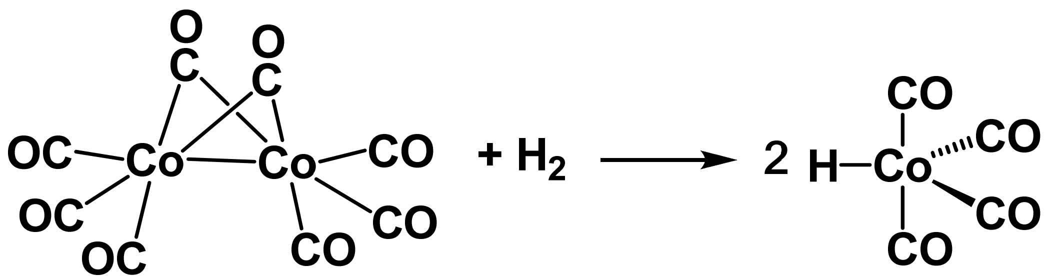
(i) Radical mechanisms

Q1. What kinds of substrates may undergo O.A. via radical mechanism?

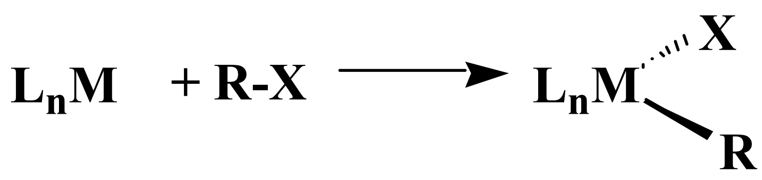
weak X-Y bonds, radical is stable

- **X_2: Cl_2, Br_2, I_2**
- **R-X: Ar-X, CH_3X, $PhCH_2X$, CH_2=CH-CH_2X**
- **Binuclear oxidative addition (usually 1e process).**

e.g.



Major difference between free radical and SN_2 mechanisms



	SN_2	Free radical
Stereochemistry at carbon	inverted	lost
Relative reactivity: RCl, RBr, RI, ROTs	R-OTs > R-I >R-Br > R-Cl	R-I > R-Br >R-Cl>R-OTs
Reactivity of R-X: MeX, 1°, 2°, 3°	MeX>1°>2°>3° (steric)	MeX<1°<2°<3° (radical stability)

Practice Problems

Oxidative addition (O. A.) to compound **A** is found to take place with MeOTs, but not with *i*-PrI. Complex **B** reacts with *i*-PrI, but not with MeOTs.

(a) What is the mechanism in each case?

(b) Which of the complexes, **A** or **B**, will be more likely to react with MeI?

(a) MeOTs is good for SN_2, but not good for radical reactions

i-PrI is not good for SN_2, but good for radical reactions

Therefore: Compound **A** undergo SN_2 O.A.

Compound **B** undergo radical O.A.

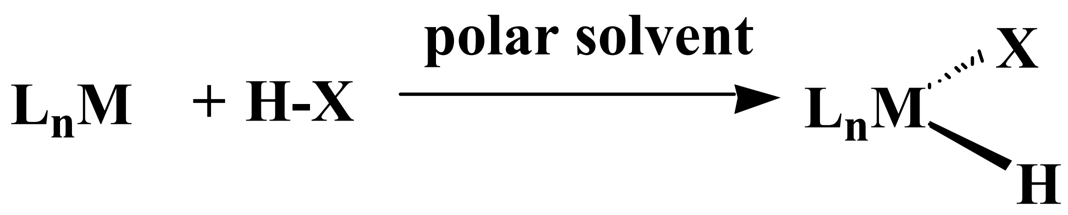
(b) Since CH_3I is good for SN_2 O. A., compound ***A*** *will react with it.*

ii) Ionic mechanism

Some molecules (e.g. HCl, HBr, HI) dissociate into ions in polar solvents.

$HCl + H_2O$ --------> H^+(aqua) + Cl^- (aqua)

Therefore oxidative addition of these molecules in polar solvents must involve ionic species.

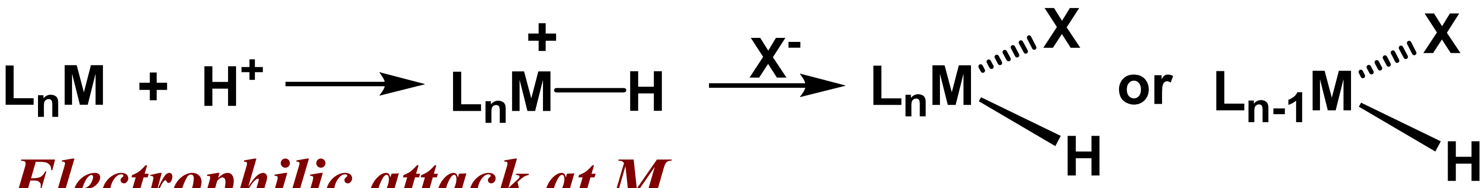


There are two possible pathways, depending on electron-richness of metal complexes

ii) Ionic mechanism

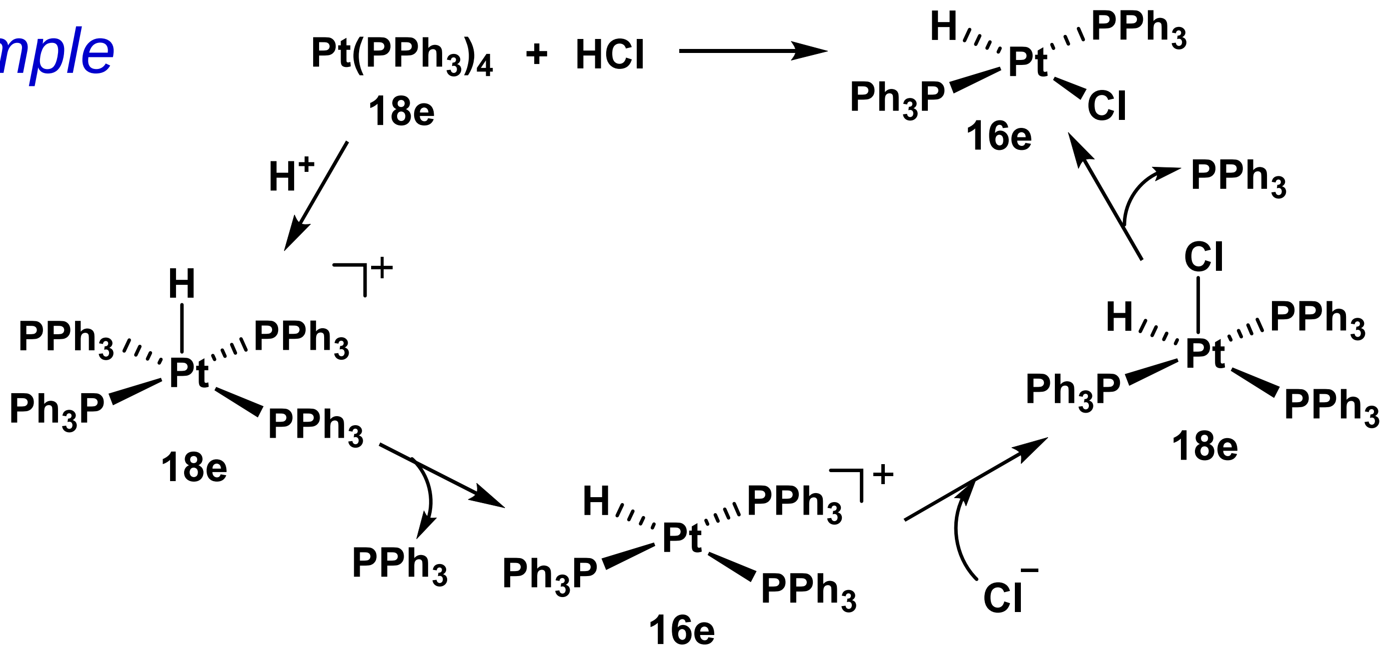
H-X $\longrightarrow$ $H^+ + X^-$ L_nM first reacts with **H^+ or X^- ?**

If L_nM is electron rich, the following sequence is likely:



Electrophilic attack at M

Example



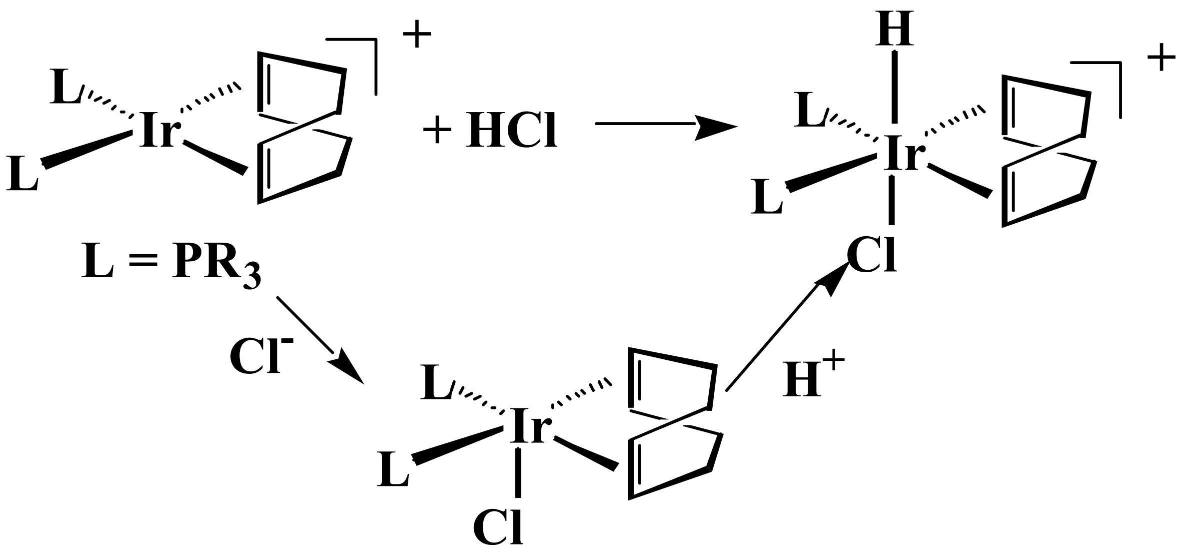
ii) Ionic mechanism

If L_nM is electron poor, the following sequence is likely:

$$H\text{-}X \longrightarrow H^+ + X^-$$

$$L_nM + X^- \longrightarrow L_n\overset{-}{M}\text{—}X \xrightarrow{H^+} L_nM(X)(H)$$

Example



C. Examples of addition of specific molecules

1). R-X bonds

2). C-H bonds

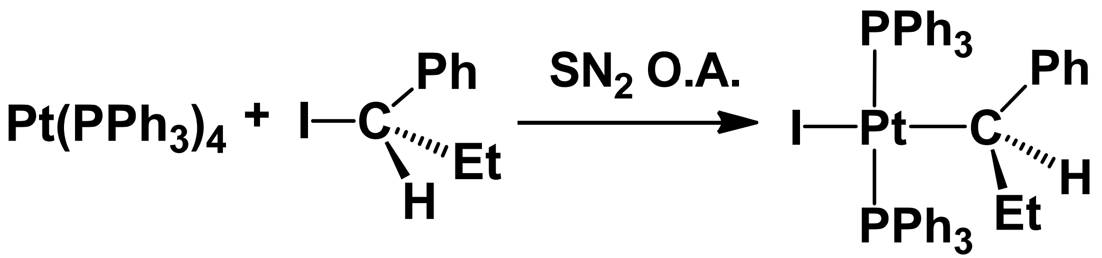
3). C-C bonds

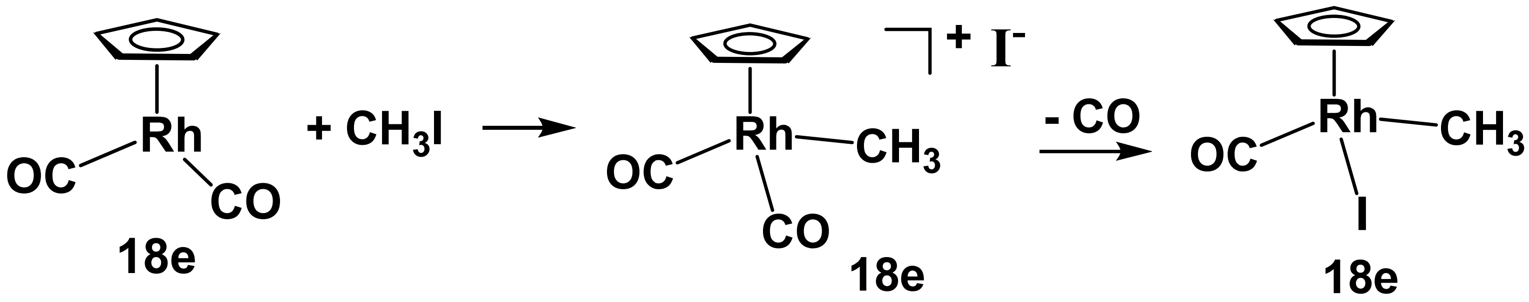
4). Other bonds

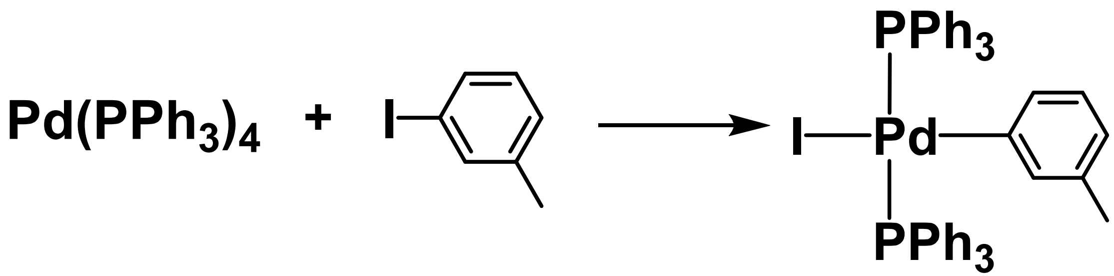
C. Examples of addition of specific molecules

1) R-X bonds

Q1. Predict the product for the following reactions.



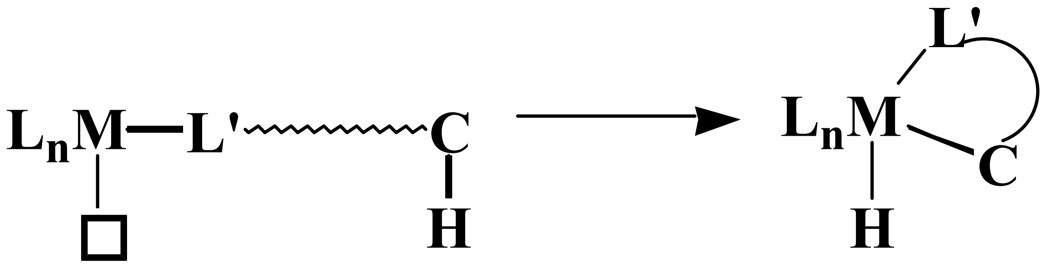




C. Examples of addition of specific molecules

2) C-H bonds

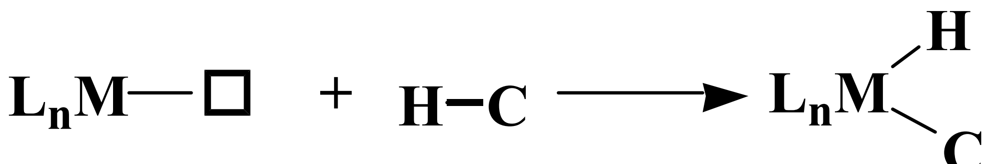
Intramolecularly, relatively easy



"cyclometallation"

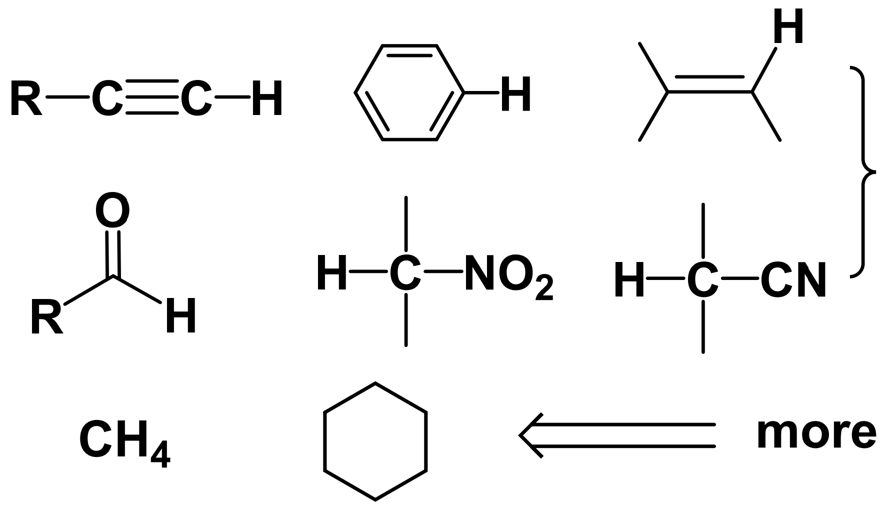
Intermoleclar, more difficult.

Need very electron-rich L_nM



16e

Examples of substrates that undergo C-H oxidative addition

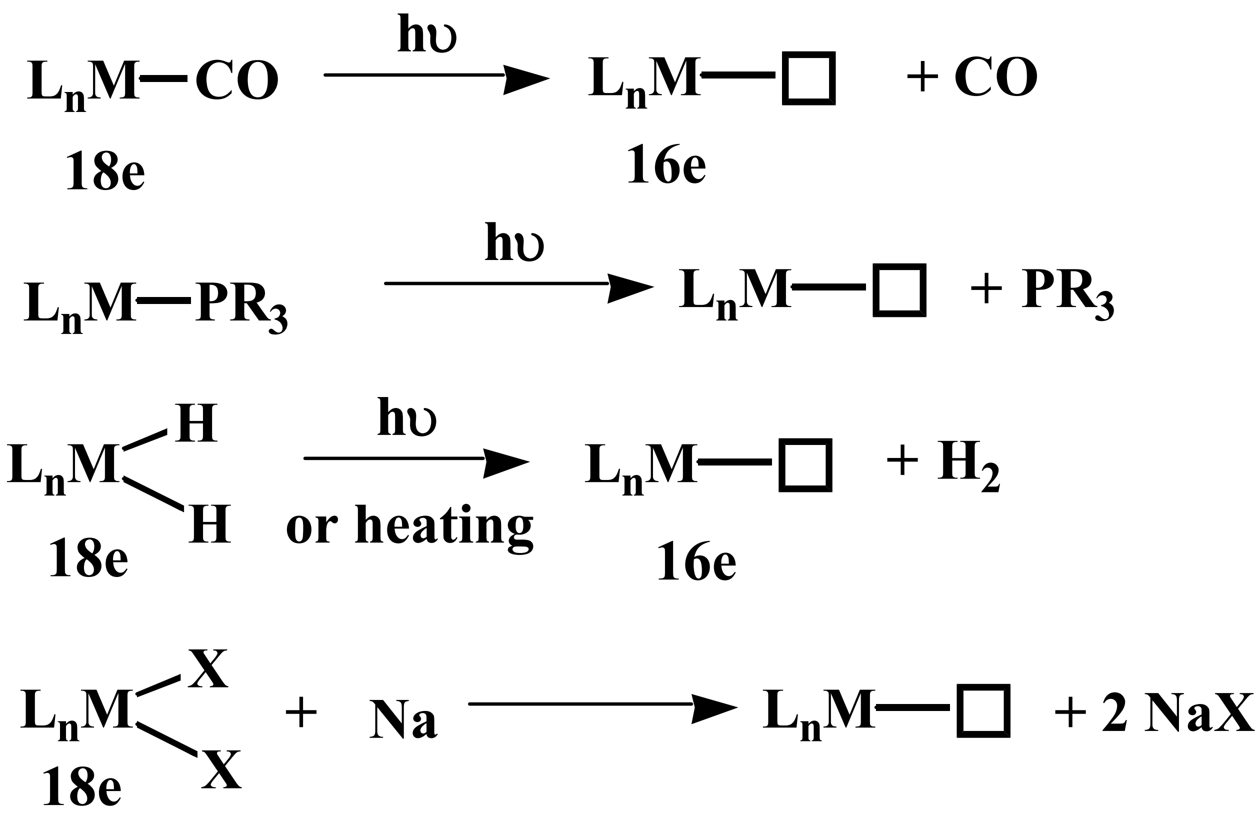


relatively easy

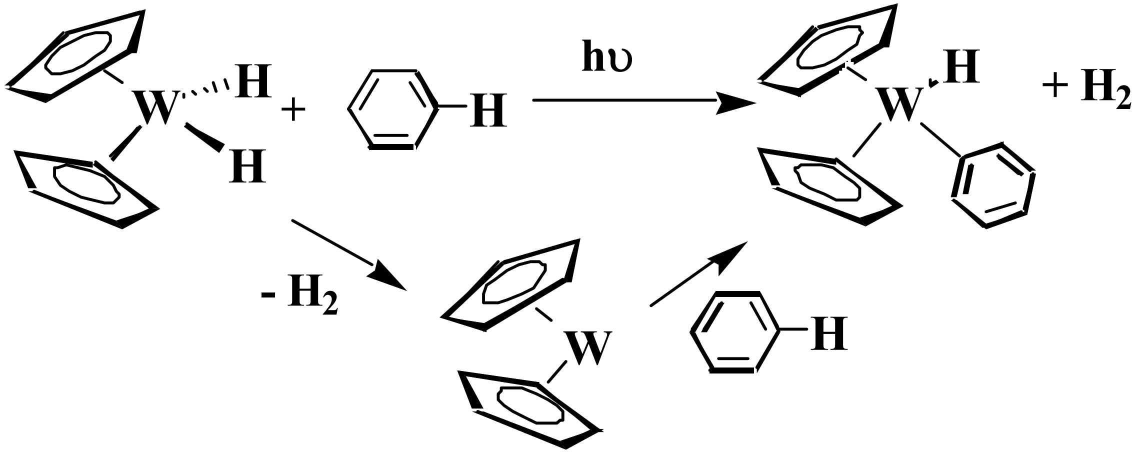
more difficult

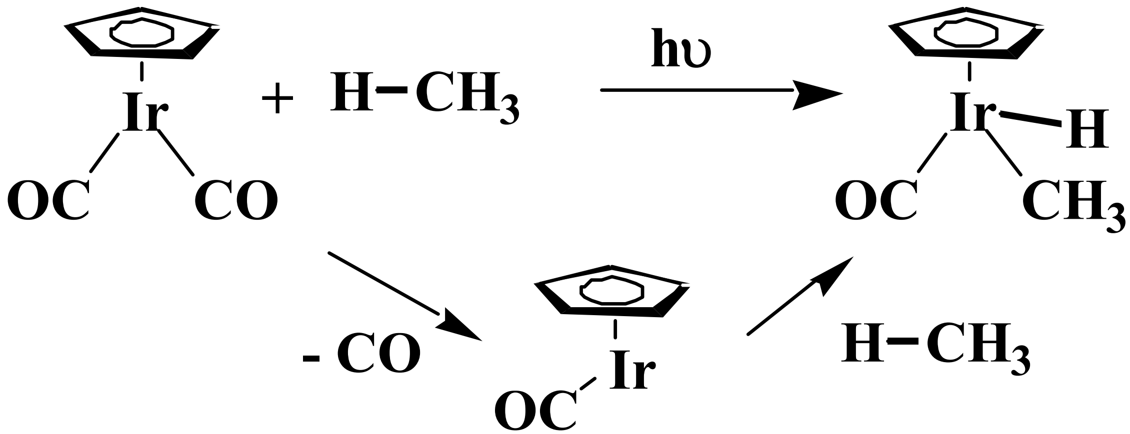
How to generate reactive 16e L_nM ?

Many ways. *e.g.*

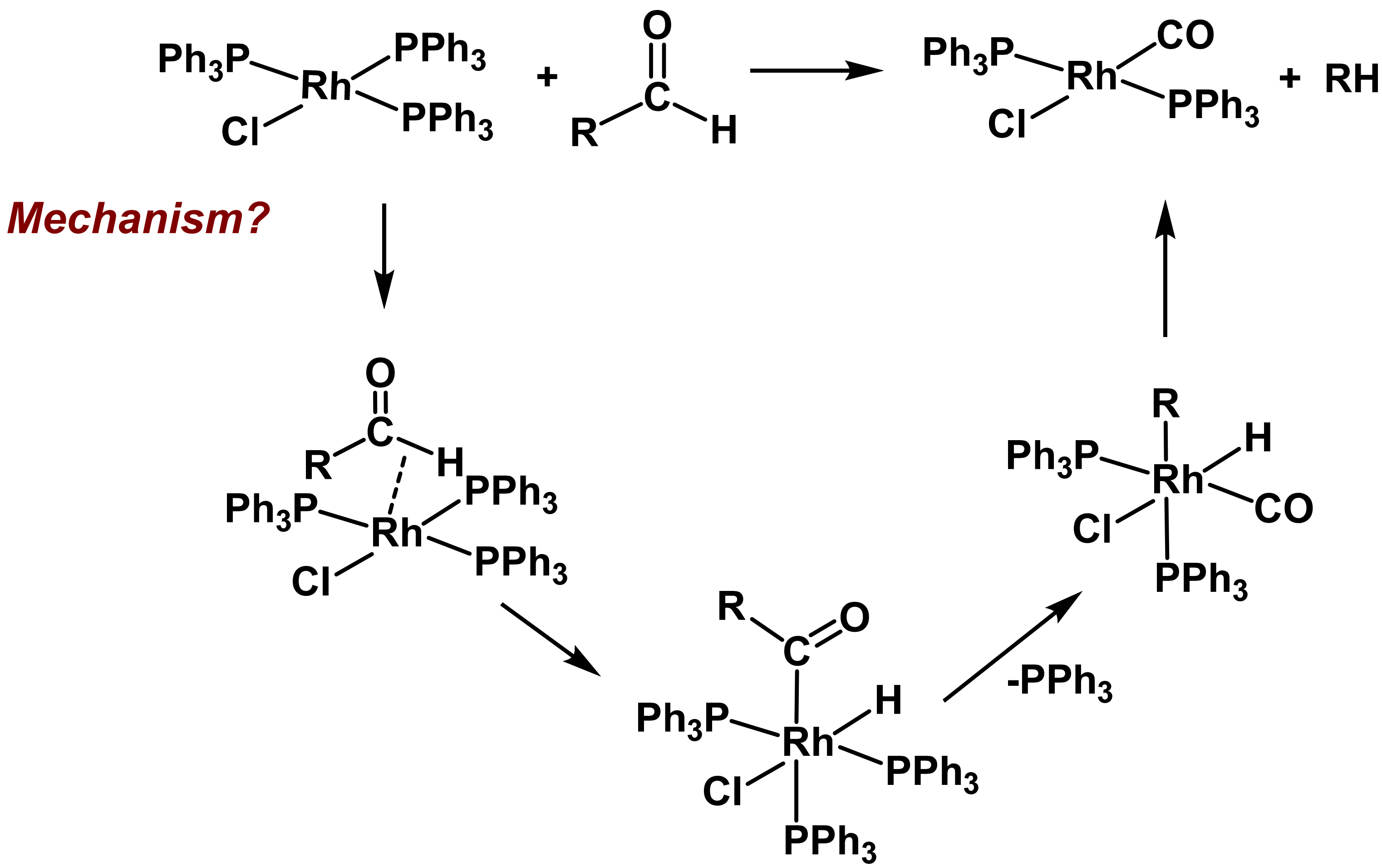


Examples of C-H bond oxidative addition reactions

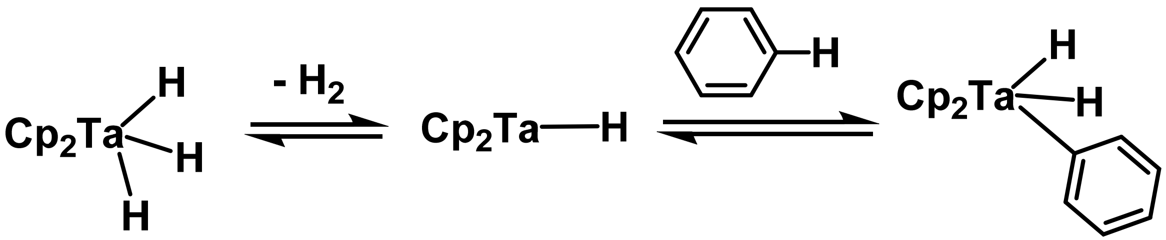




C-H bond activation is involved in many reactions. e.g.

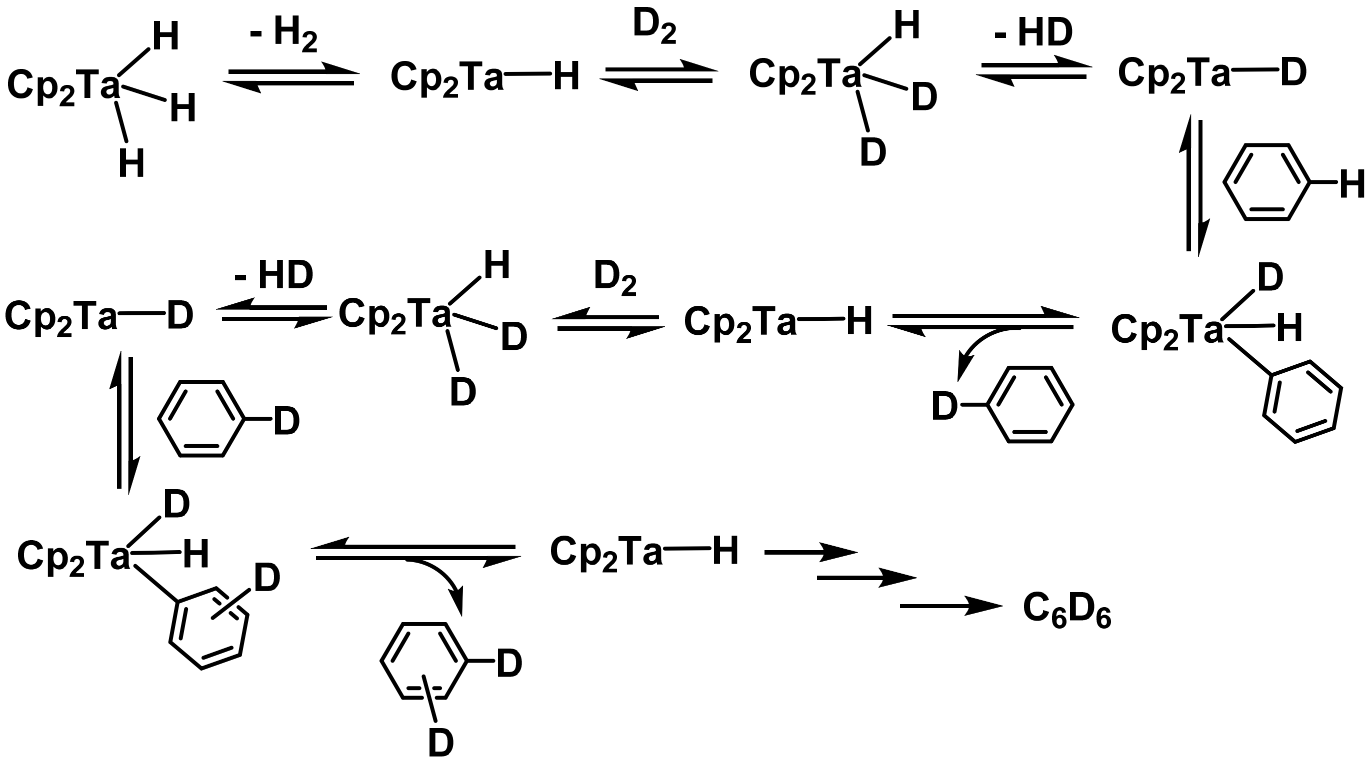


C-H bond activation is involved in many reactions. e.g.



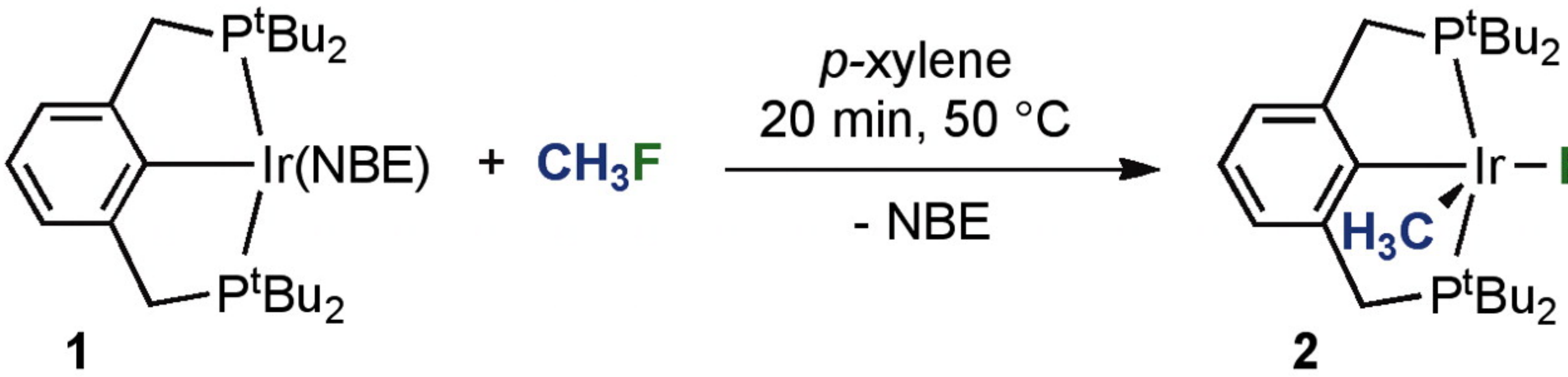
$$D_2 + C_6H_6 \xrightarrow[\text{catalytic amount}]{Cp_2TaH_3} HD + C_6D_6$$

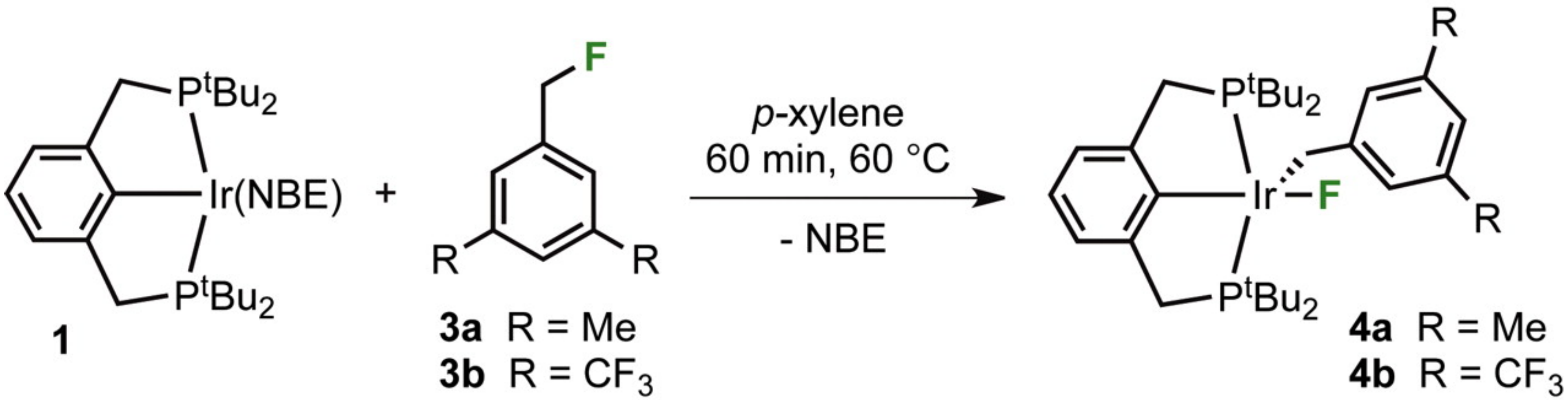
Mechanism?



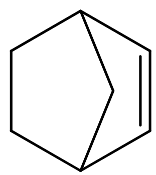
J Choi, *et al. Science,* 2011, *332*, 1545-1548

Net Oxidative Addition of C(sp³)-F Bonds to Iridium via Initial C-H Bond Activation





NBE: norbornene
(降冰片烯)



Net Oxidative Addition of C(sp^3)-F Bonds to Iridium *via* Initial C-H Bond Activation

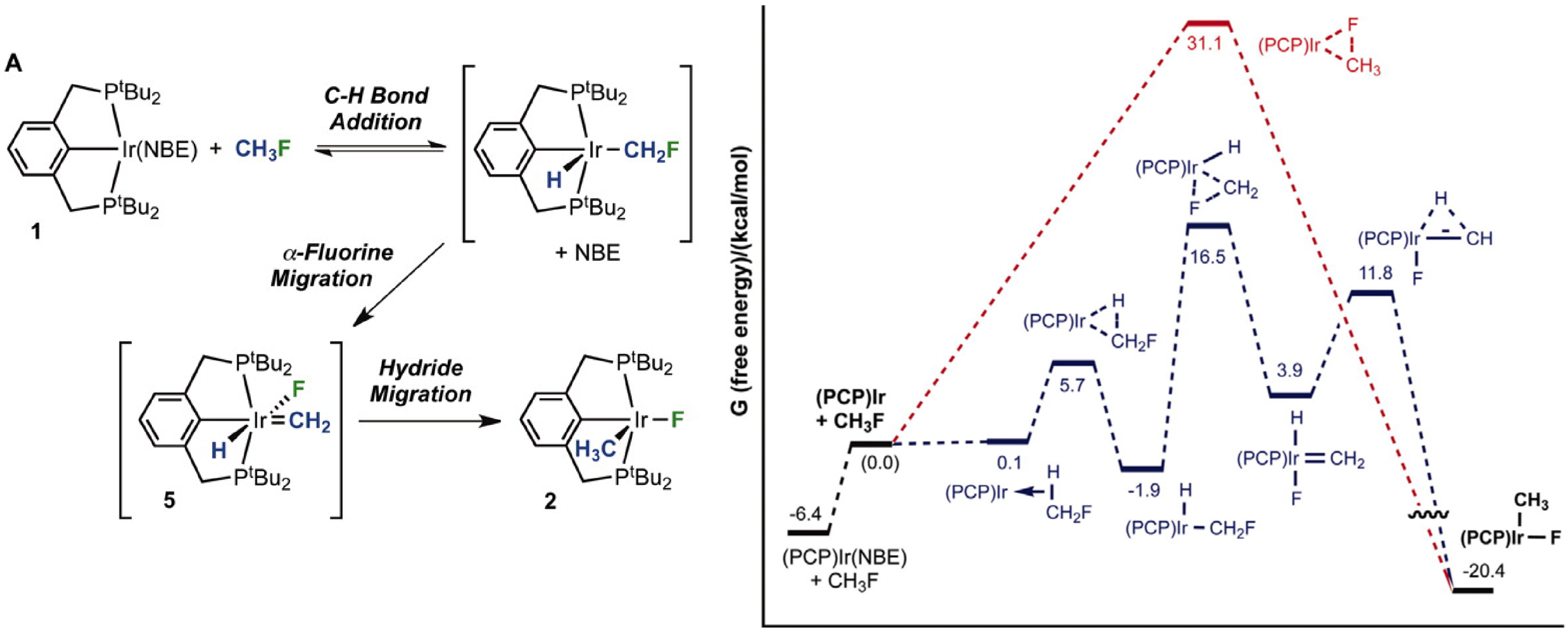
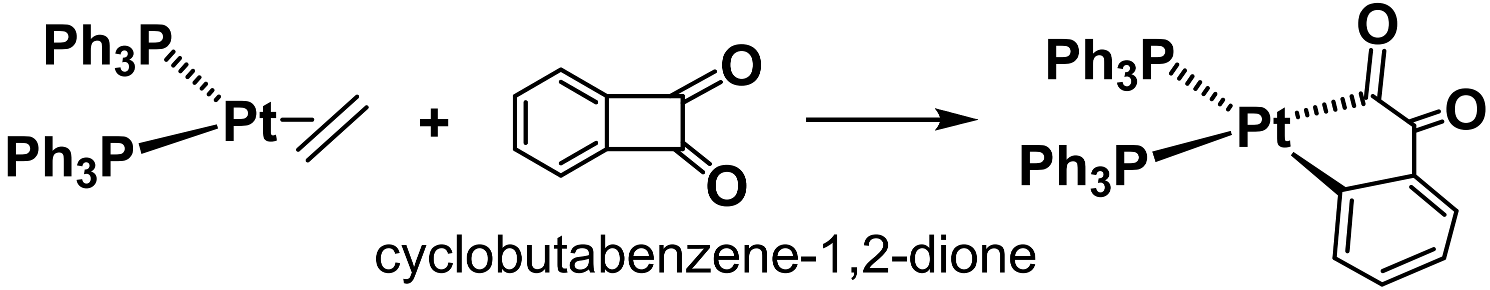


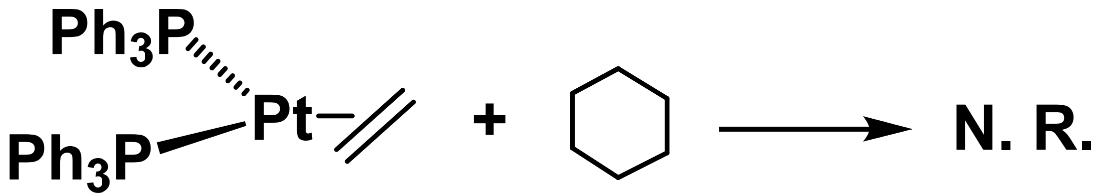
Fig. 4 (A) Proposed mechanism for oxidative addition of CH_3-F to (PCP)Ir. (B) Free energy diagrams pertaining to the proposed pathway for the reaction of (PCP)Ir with CH_3F (blue) and to direct C–F addition (red).

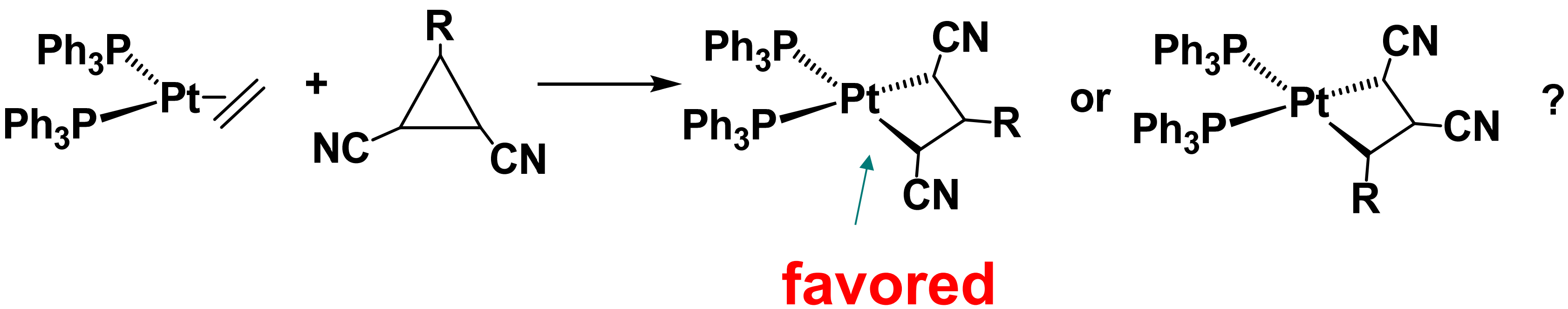
J. Choi, *et al. Science*, **2011**, *332*, 1545-1548.

C. Examples of addition of specific molecules

3) C-C bonds



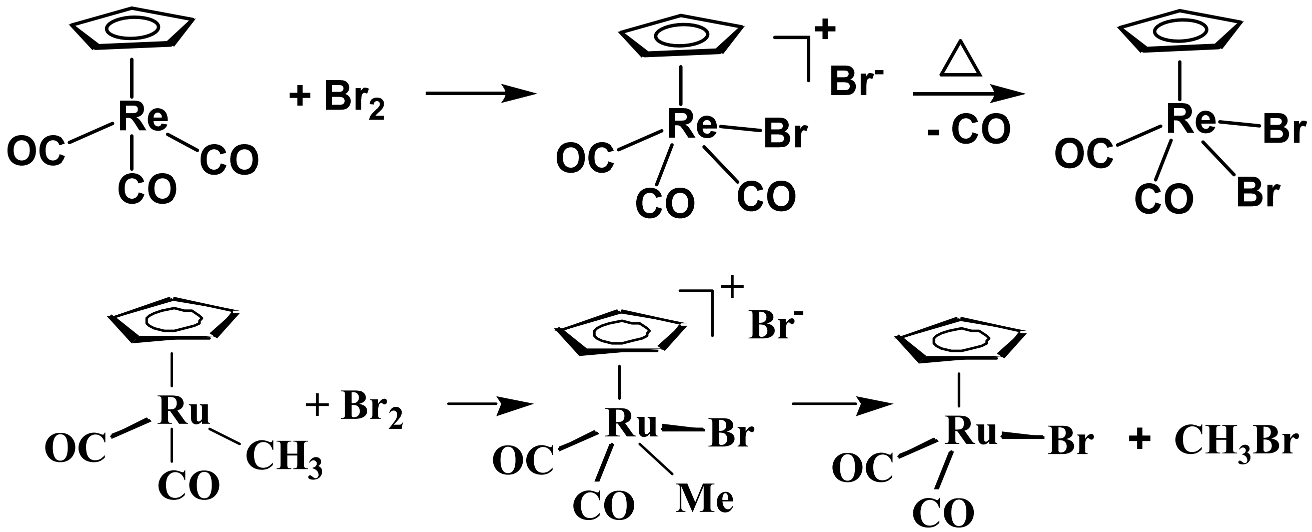




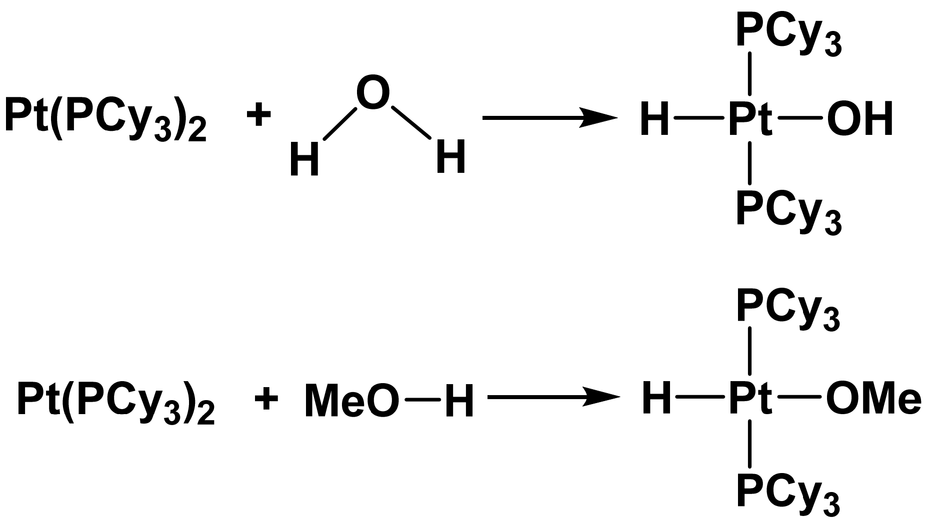
C. Examples of addition of specific molecules

4) Other bonds

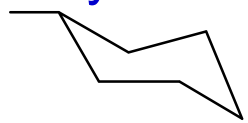
X-X



O-H



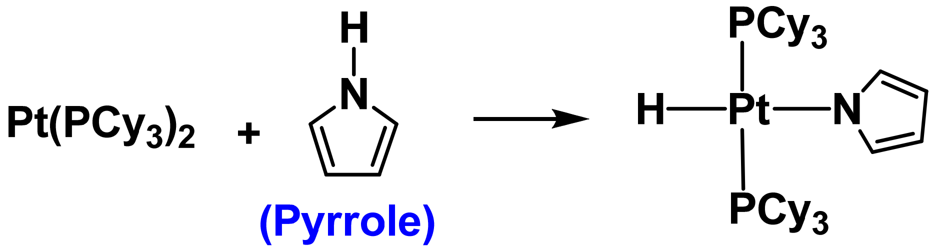
Cy = Cyclohexanyl

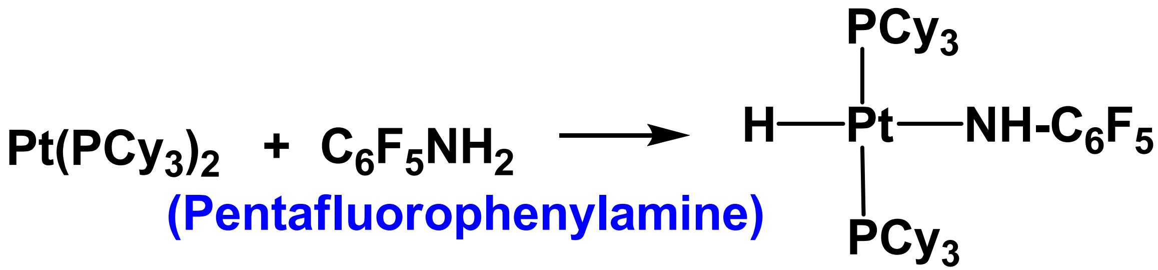


C. Examples of addition of specific molecules

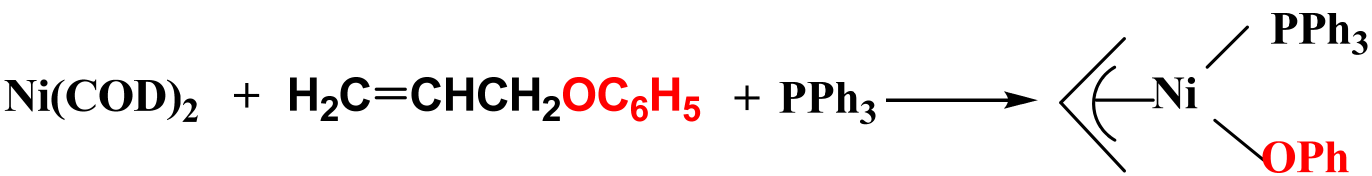
4) Other bonds

N-H

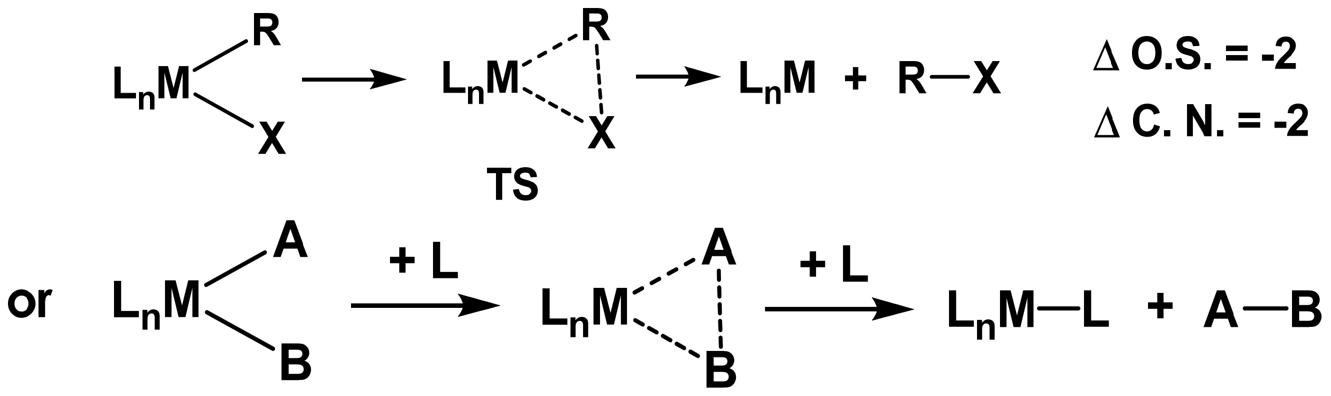




C-O bonds



D. Reductive elimination

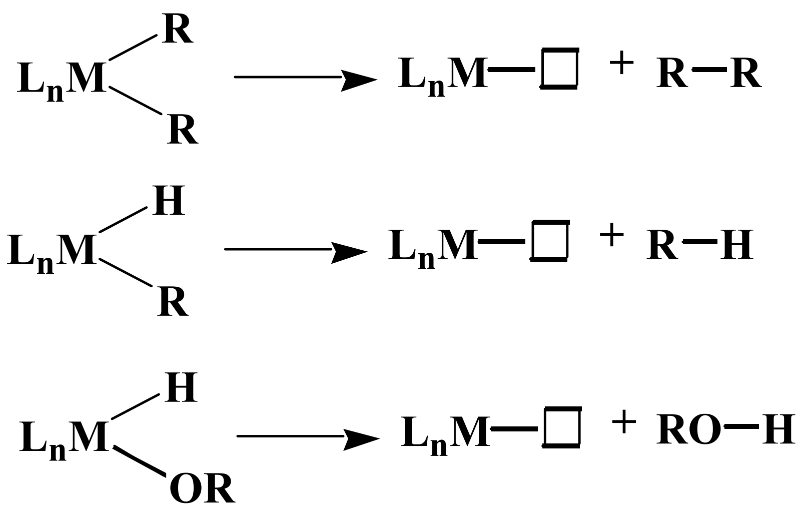


Δ O.S. = -2

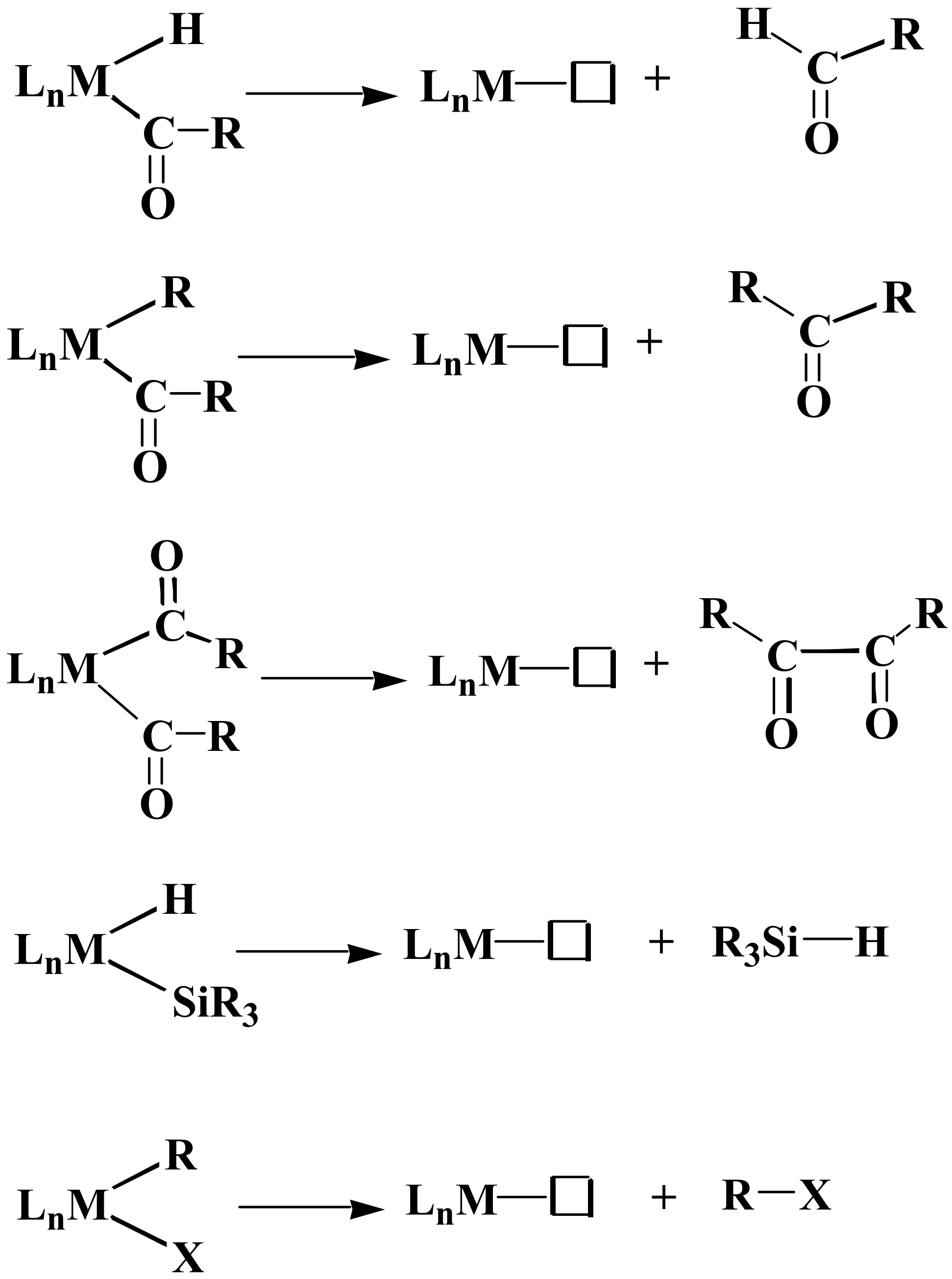
Δ C. N. = -2

Important rxn in catalysis, last step

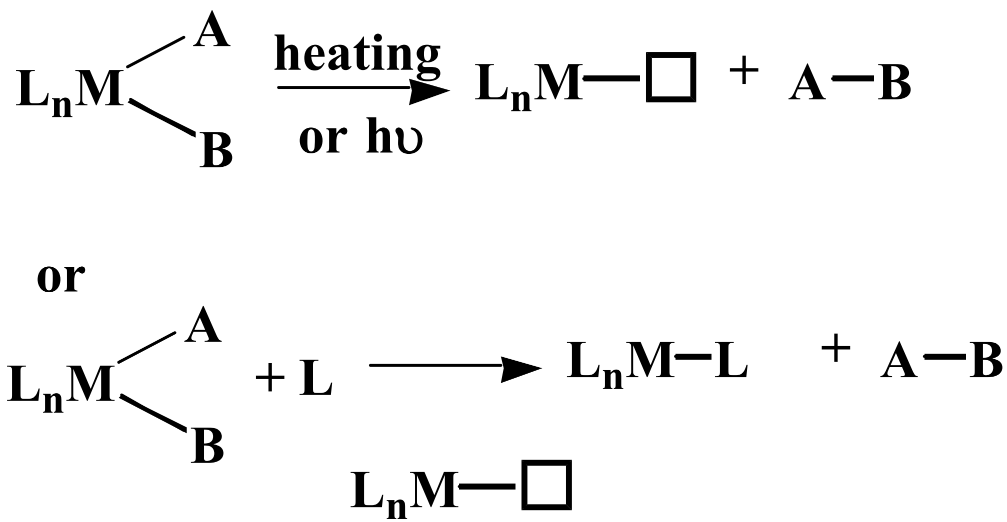
Examples:

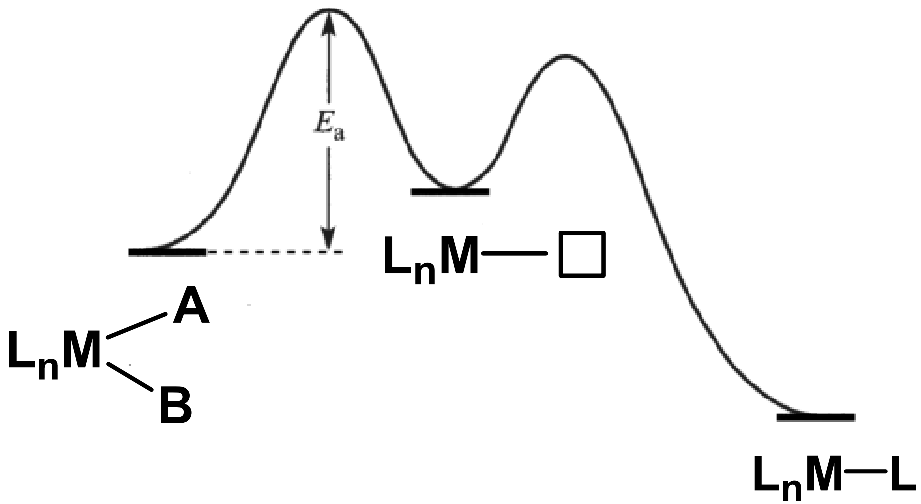


Additional Examples:

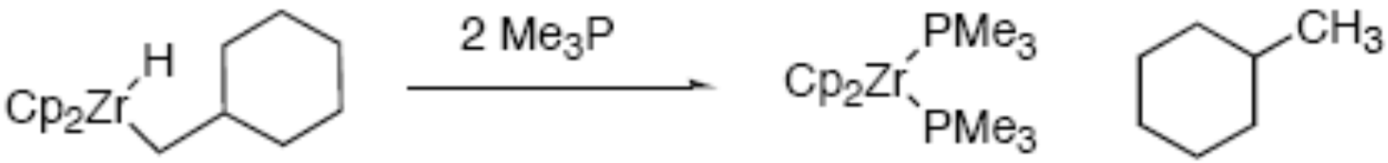


Typical reaction conditions for RE

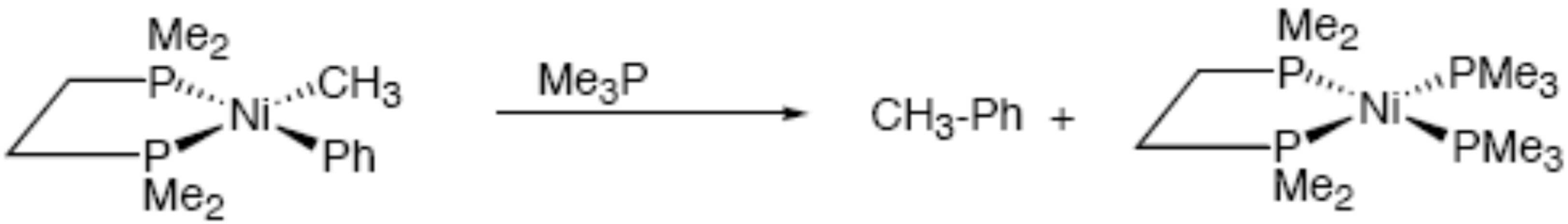




Examples:



Schwartz, *J. Am. Chem. Soc.* **1981**, *103*, 2687-2695



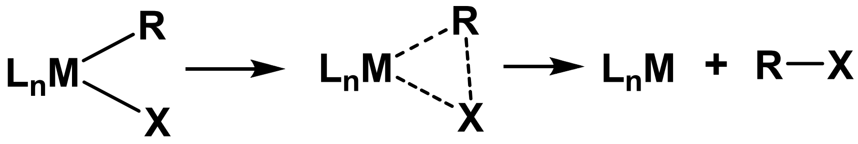
Tatsumi, *J. Am. Chem. Soc.*, **1984**, *106*, 8181 (theoretical treatment)

Detailed mechanism of reductive elimination?

Could be very complicated.

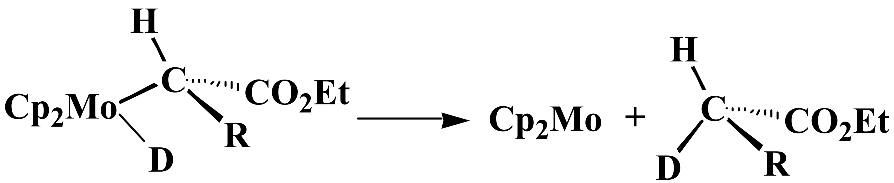
The following seems true for mononuclear complexes.

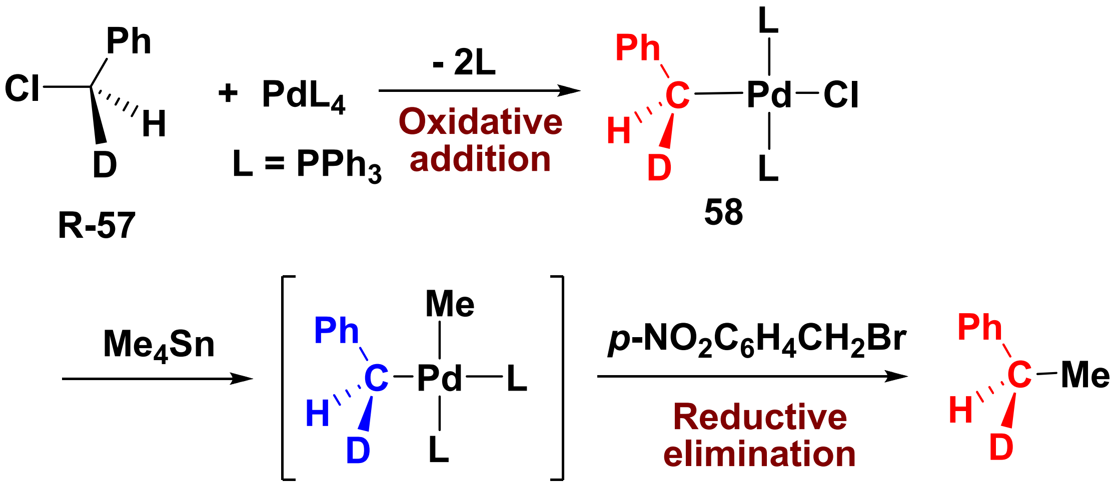
1). Transition state:



TS

2) Stereochemistry at R is retained.





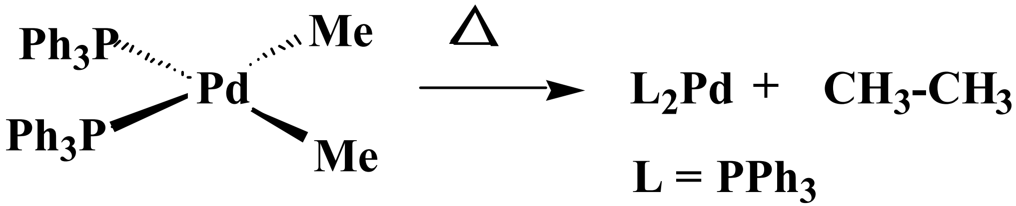
R-57

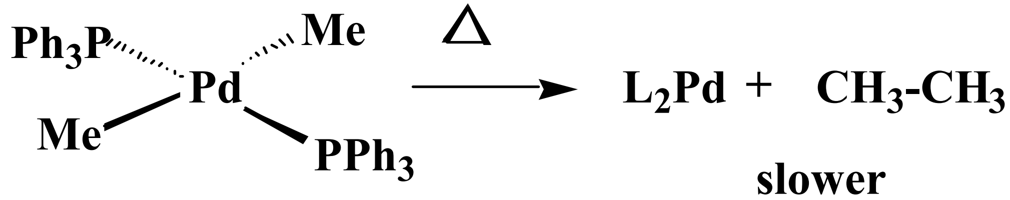
58

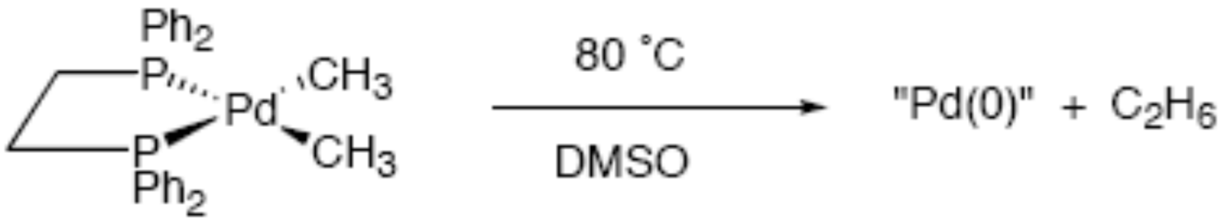
S-59

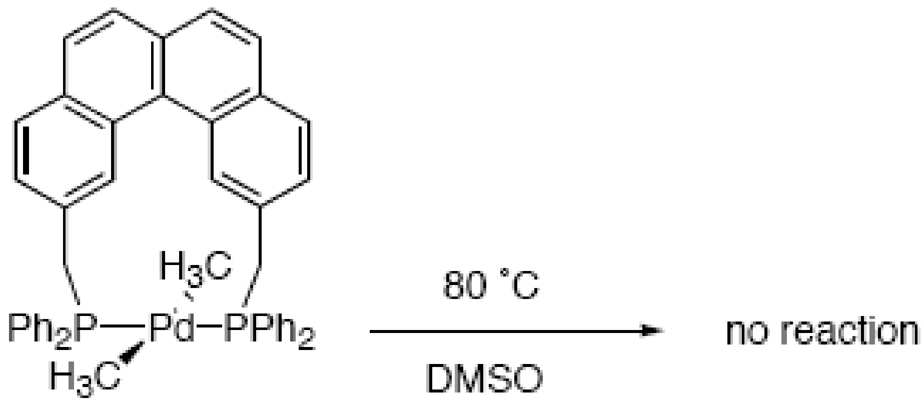
R-60

3). The two groups which reductively eliminate **must be cis** to each other.



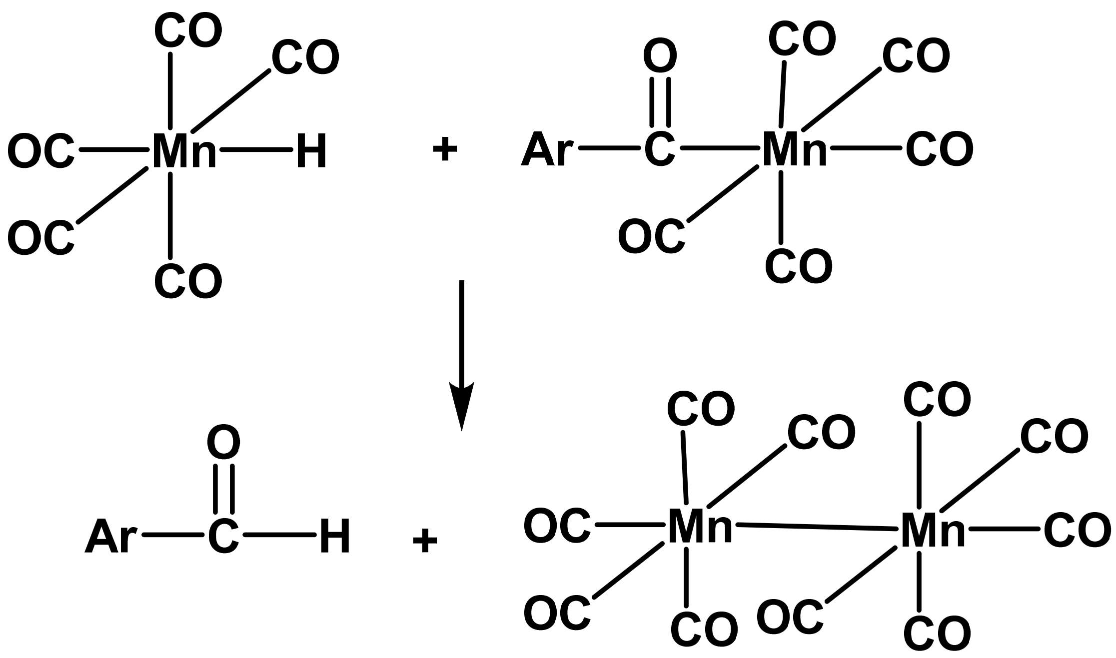




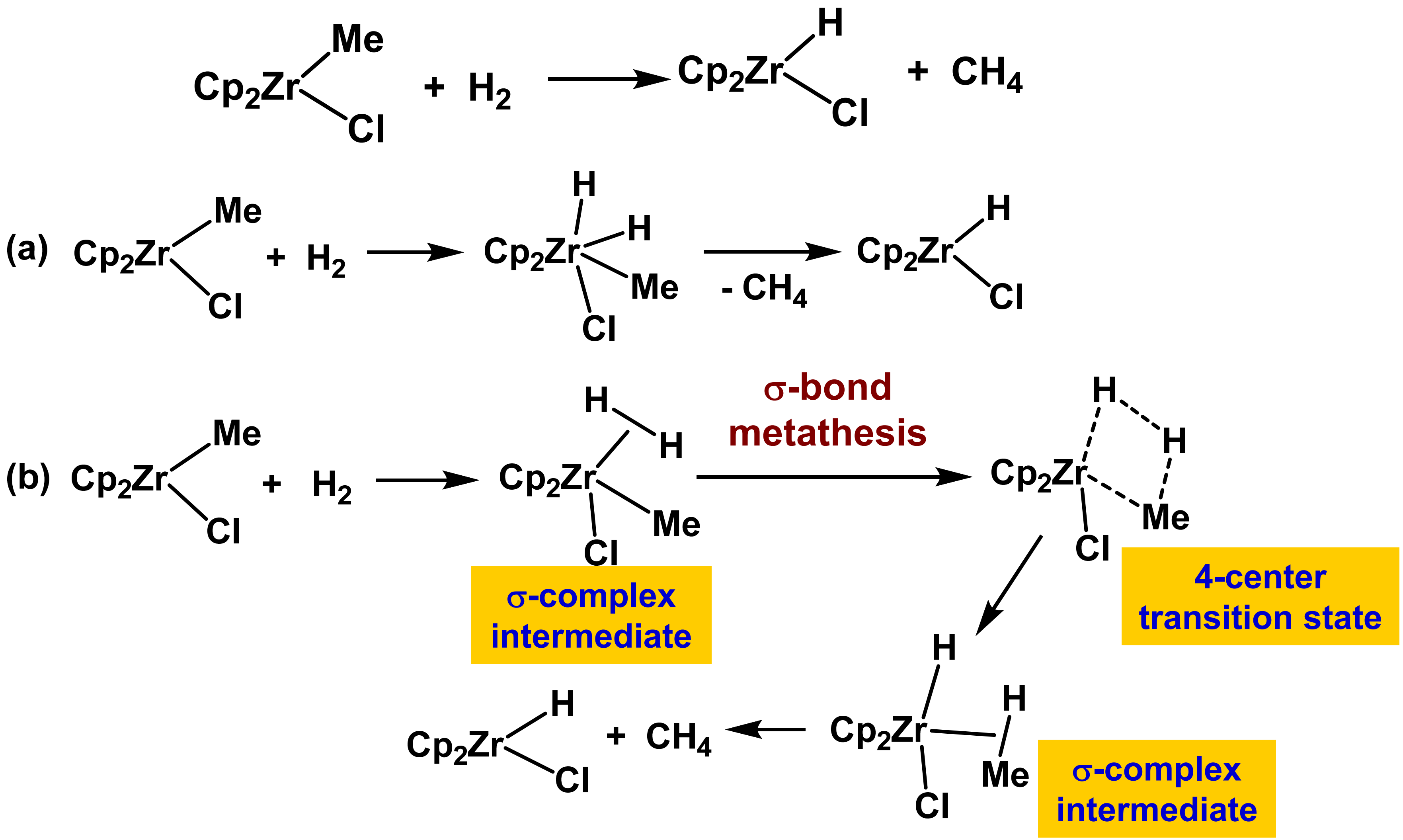


More complicated cases

1). Binuclear reductive elimination.



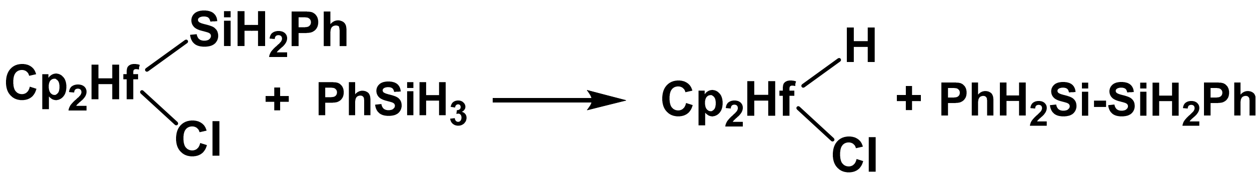
2). Apparent oxidative addition and reductive elimination reactions.（表观的/表面上的氧化加成和还原消除）



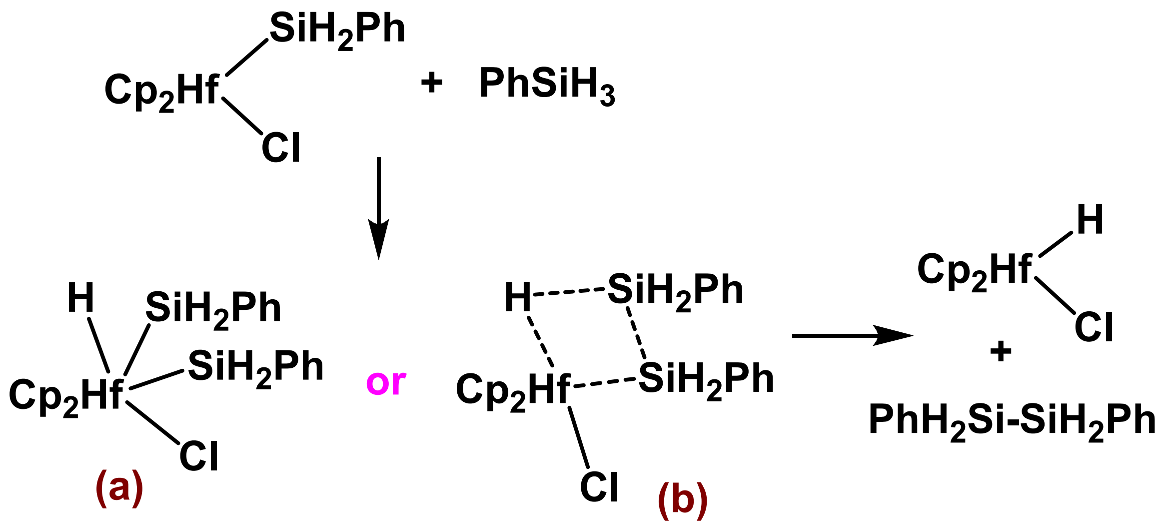
Which pathway is more likely?

(b). The O.S. of Zr in (a) is unlikely.

2). Apparent oxidative addition and reductive elimination reactions.（表观的/表面上的氧化加成和还原消除）



Possible intermediates or transition states:



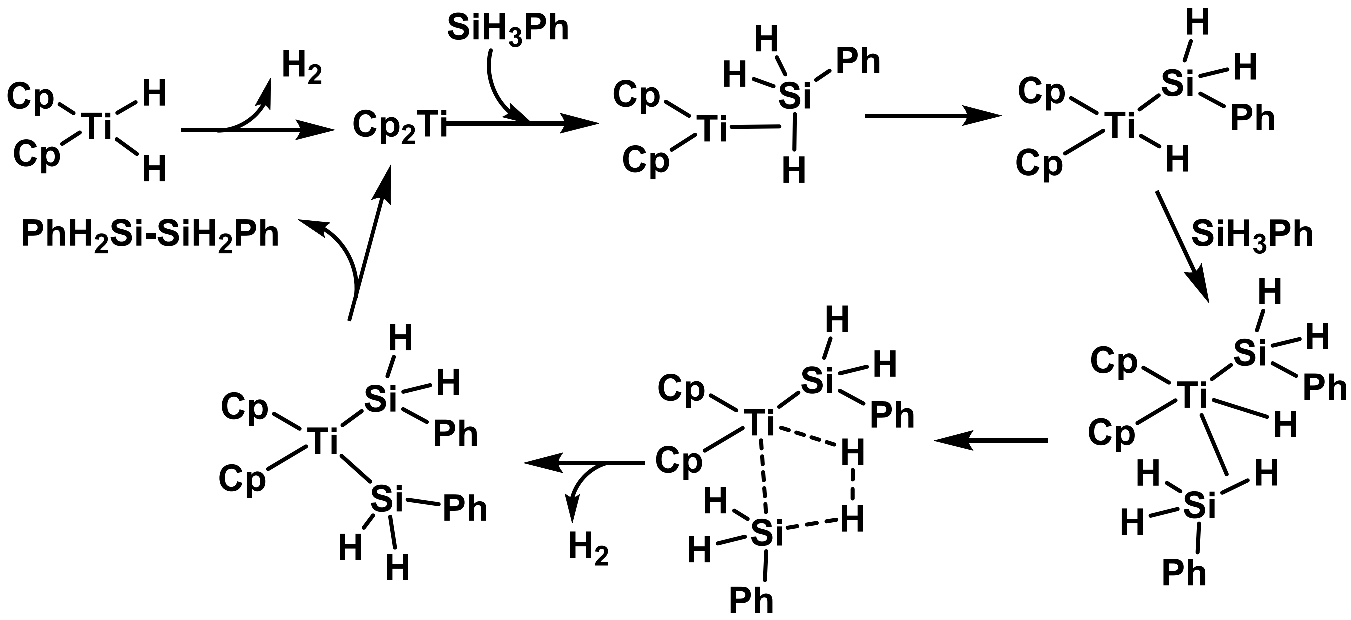
Which pathway is more likely?

Answer: (b) The O.S. of Hf in (a) is unlikely.

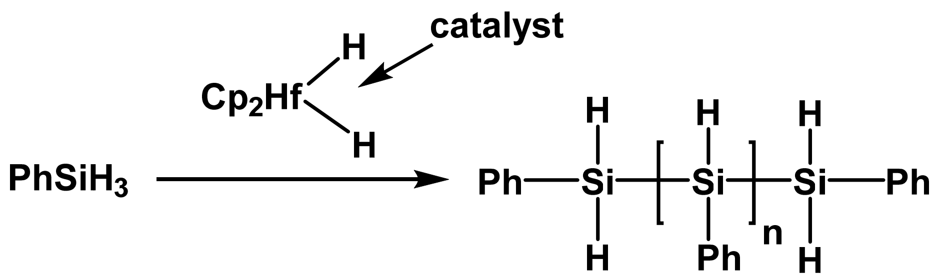
2). Apparent oxidative addition and reductive elimination reactions.（表观的/表面上的氧化加成和还原消除）

$$PhSiH_3 \xrightarrow[\text{catalytic amount}]{Cp_2TiH_2} H_2 + PhSiH_2SiH_2Ph$$

Mechanism?

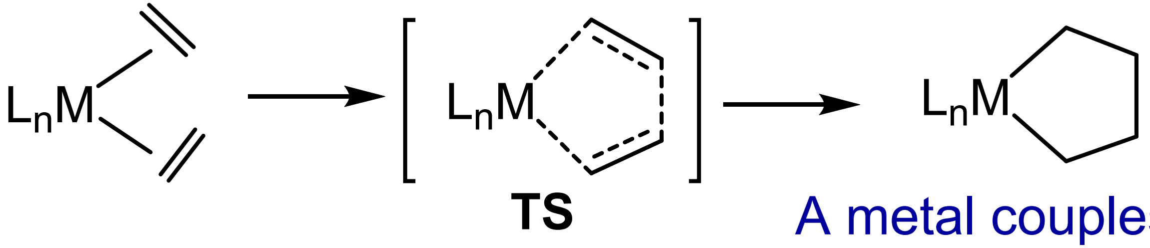


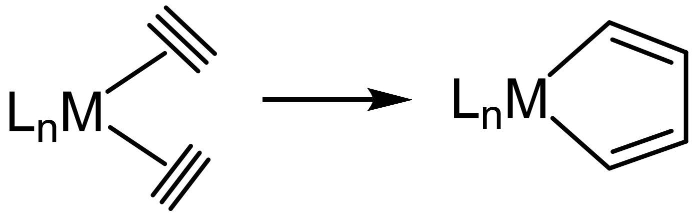
Similar mechanism can explain the following rxn.



E. Oxidative coupling and reductive fragmentation

Oxidative coupling





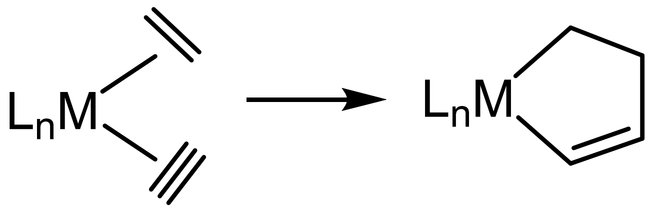
A metal couples two alkenes to give a metalacycle or two alkynes to give a *metallole.*

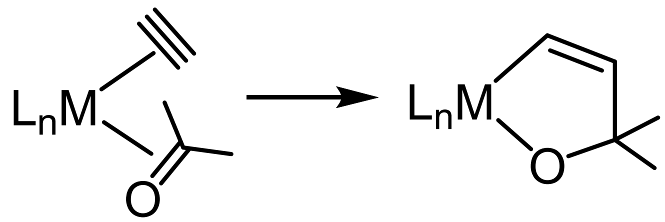
Δ O.S. = 2

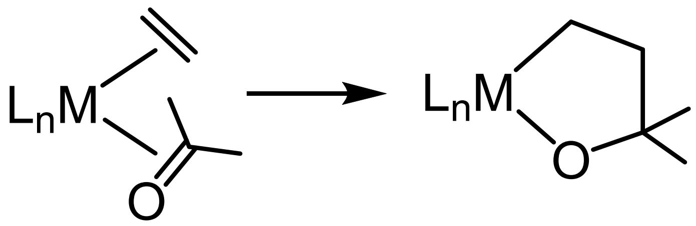
# of valence e- count = -2

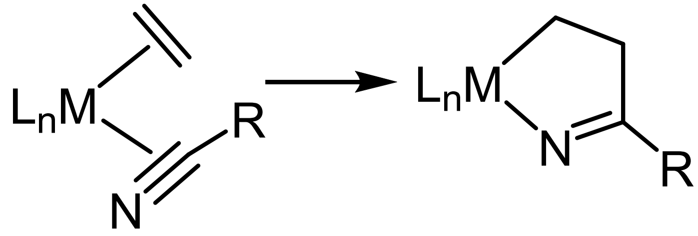
Δ C.N. = 0

Related reactions

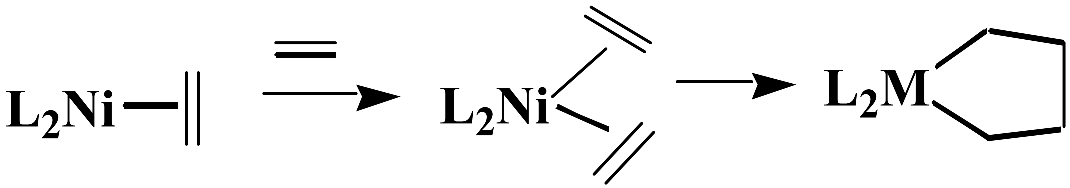




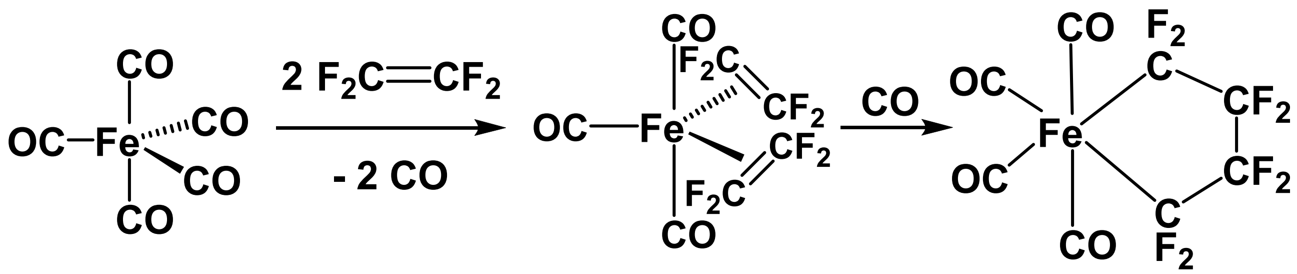


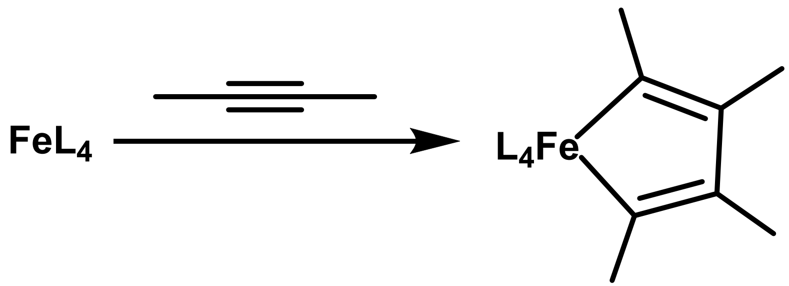


Examples



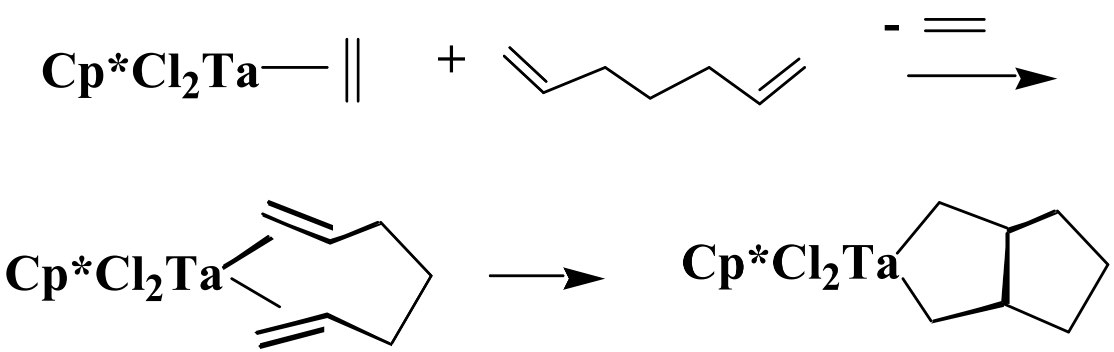
Alkynes undergo the reaction more easily than do alkenes unless activated by the substituents or by strain.

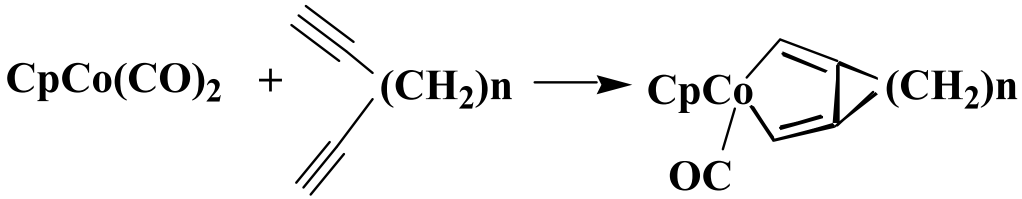


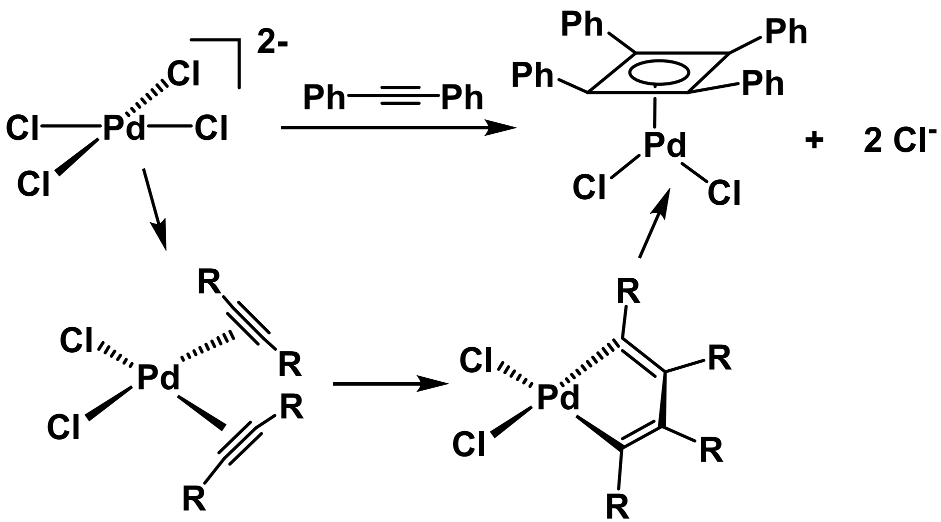


C_2F_4 undergoes the reaction easily because F prefers the *sp*^{3} C–F of the product, where the high electronegativity of F is better satisfied by the less electronegative *sp*^{3} carbon and repulsion between the F lone pairs and the C=C π-bond is relieved.

Examples







These reactions are useful in synthesis and catalysis.

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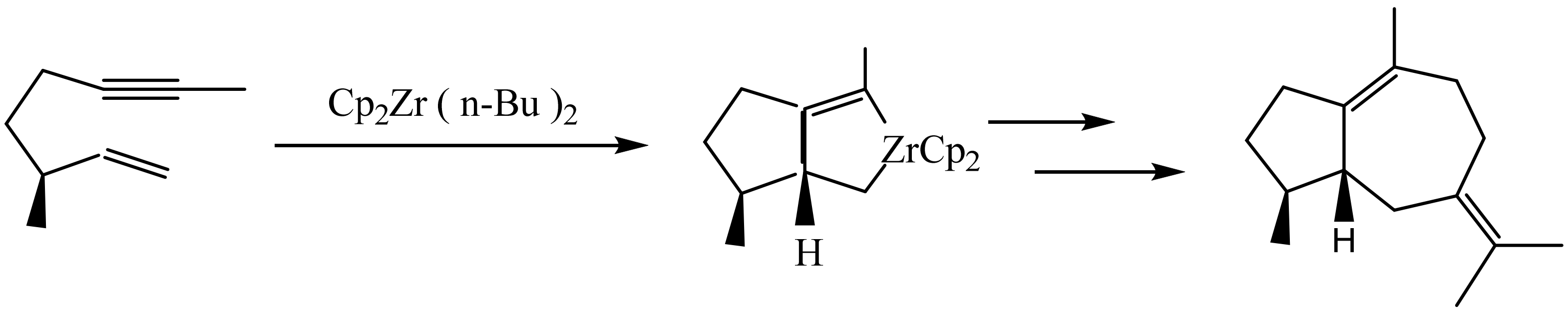
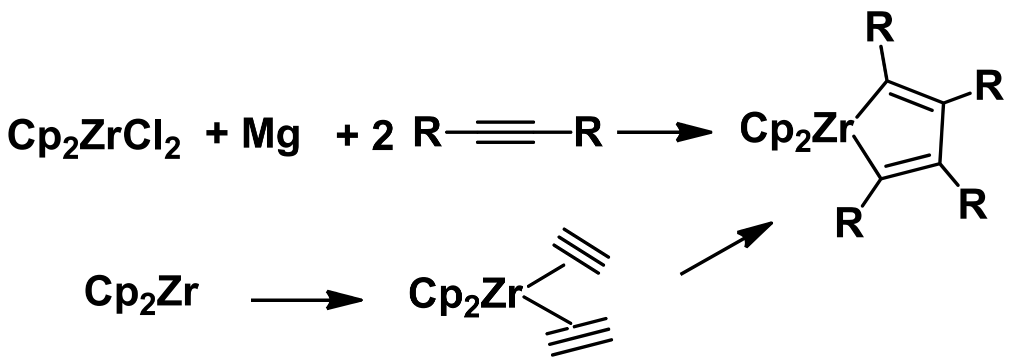
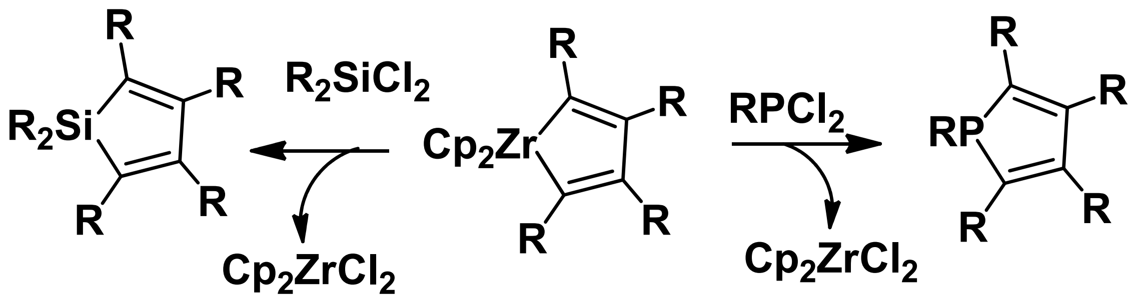


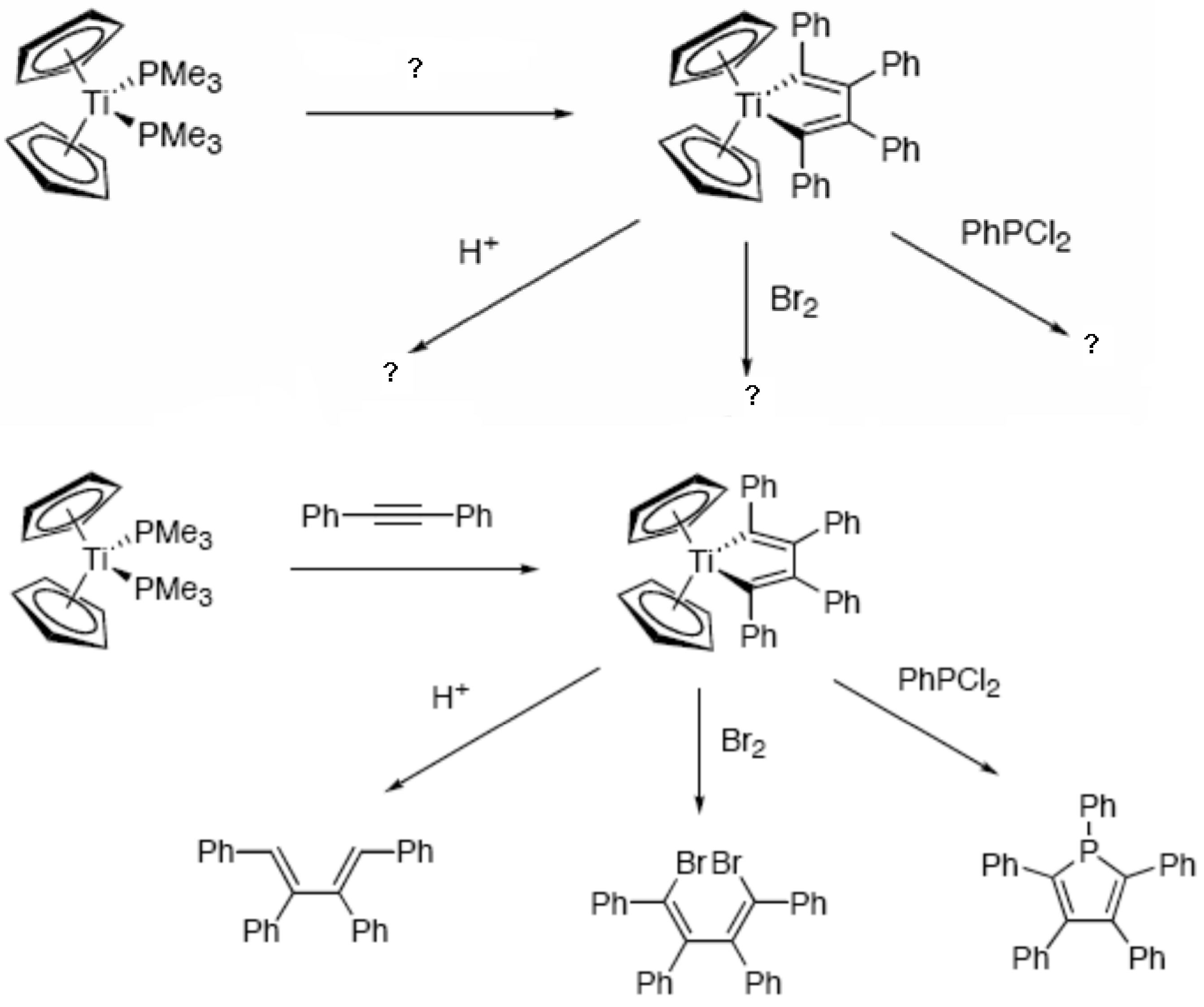
β-bulnesene





Practice Problems

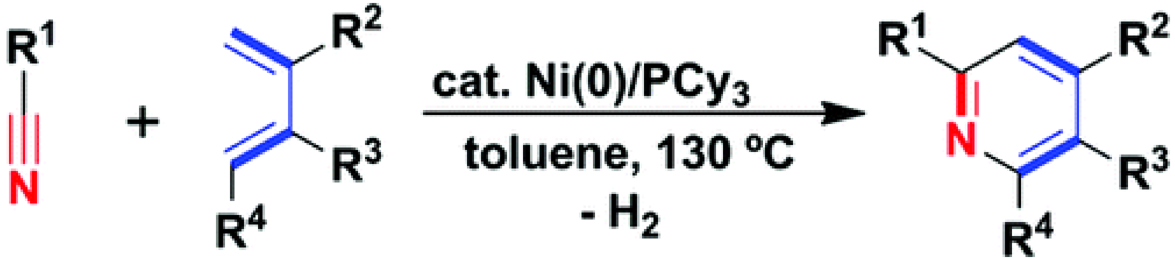
Suggest reagents or products.



Suggested reading

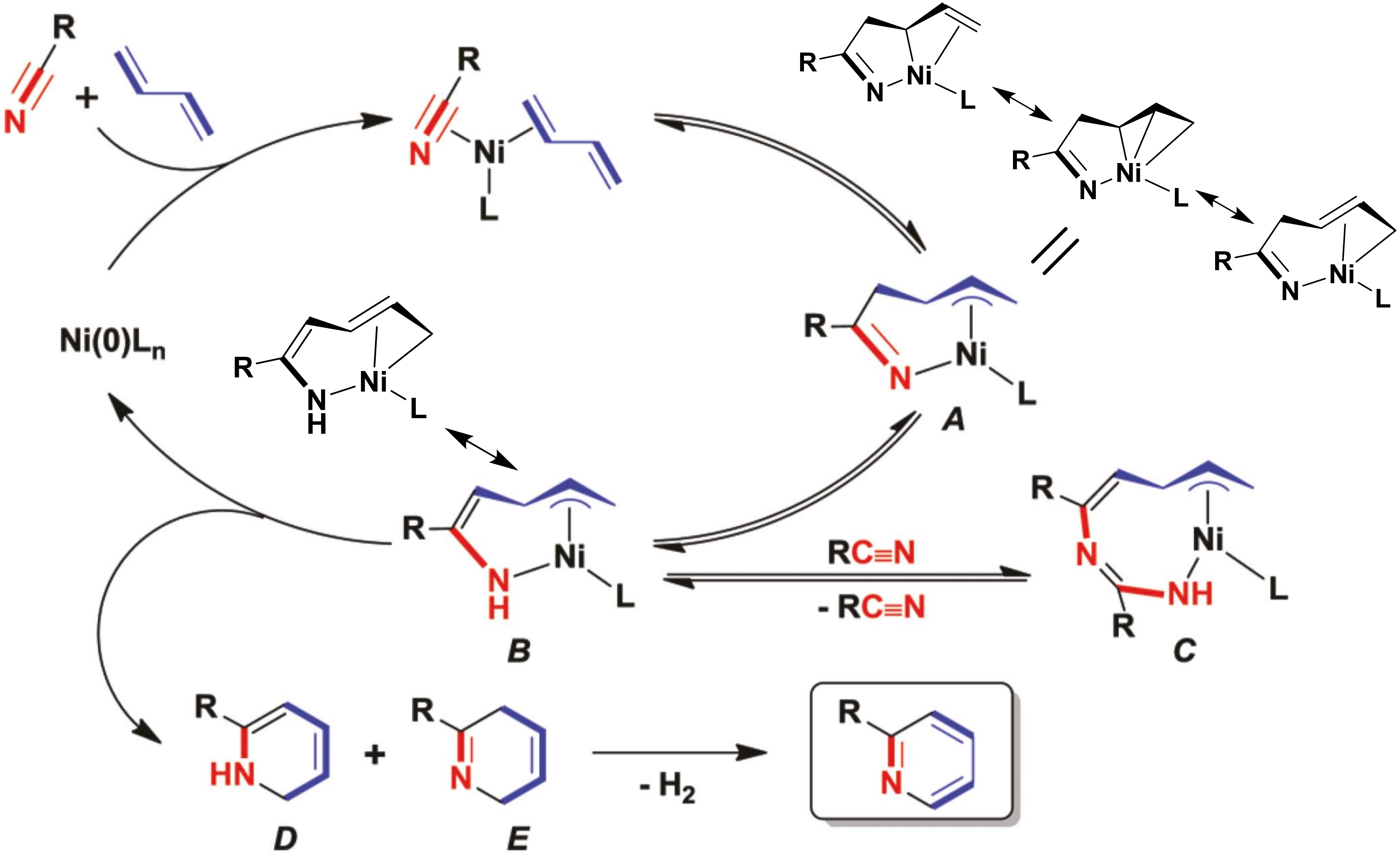
Nickel-Catalyzed Dehydrogenative [4 + 2] Cycloaddition of 1,3-Dienes with Nitriles

Masato Ohashi†‡, Ippei Takeda†, Masashi Ikawa†, and Sensuke Ogoshi*†
†Department of Applied Chemistry, Faculty of Engineering, and ‡Center for Atomic and Molecular Technologies, Osaka University, Suita, Osaka 565-0871, Japan



1. transition-metal-catalyzed [2 + 2 + 2] cycloaddition reaction of two alkynes and a nitrile is one of the most powerful methods for preparing versatile, highly substituted pyridine derivatives.
2. However, the lack of chemo- and regioselectivity is still a crucial issue associated with fully intermolecular [2 + 2 + 2] cycloaddition.
3. The present study developed the Ni(0)-catalyzed intermolecular dehydrogenative [4 + 2] cycloaddition reaction of 1,3-butadienes with nitriles to give a variety of pyridines regioselectively.

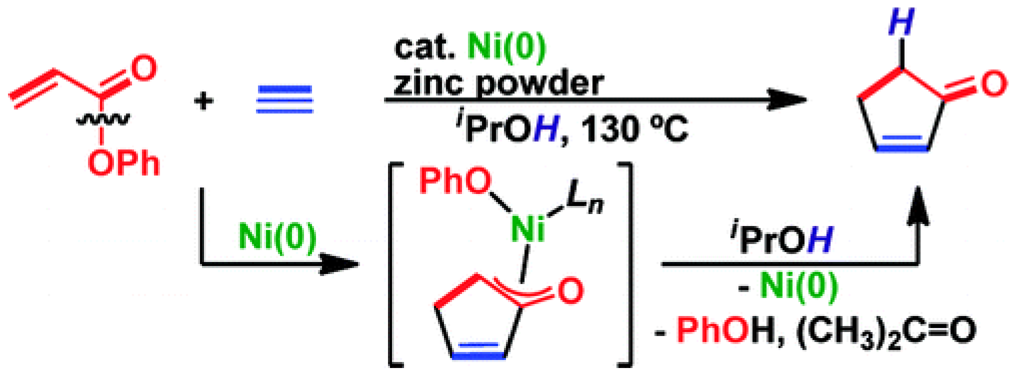
J. Am. Chem. Soc., **2011**, *133*, 18018.


Ni(0)Ln
RC≡N
- RC≡N
- H2
A
B
C
D
E

Nickel-Catalyzed Formation of Cyclopentenone Derivatives via the Unique Cycloaddition of *α,β*-Unsaturated Phenyl Esters with Alkynes

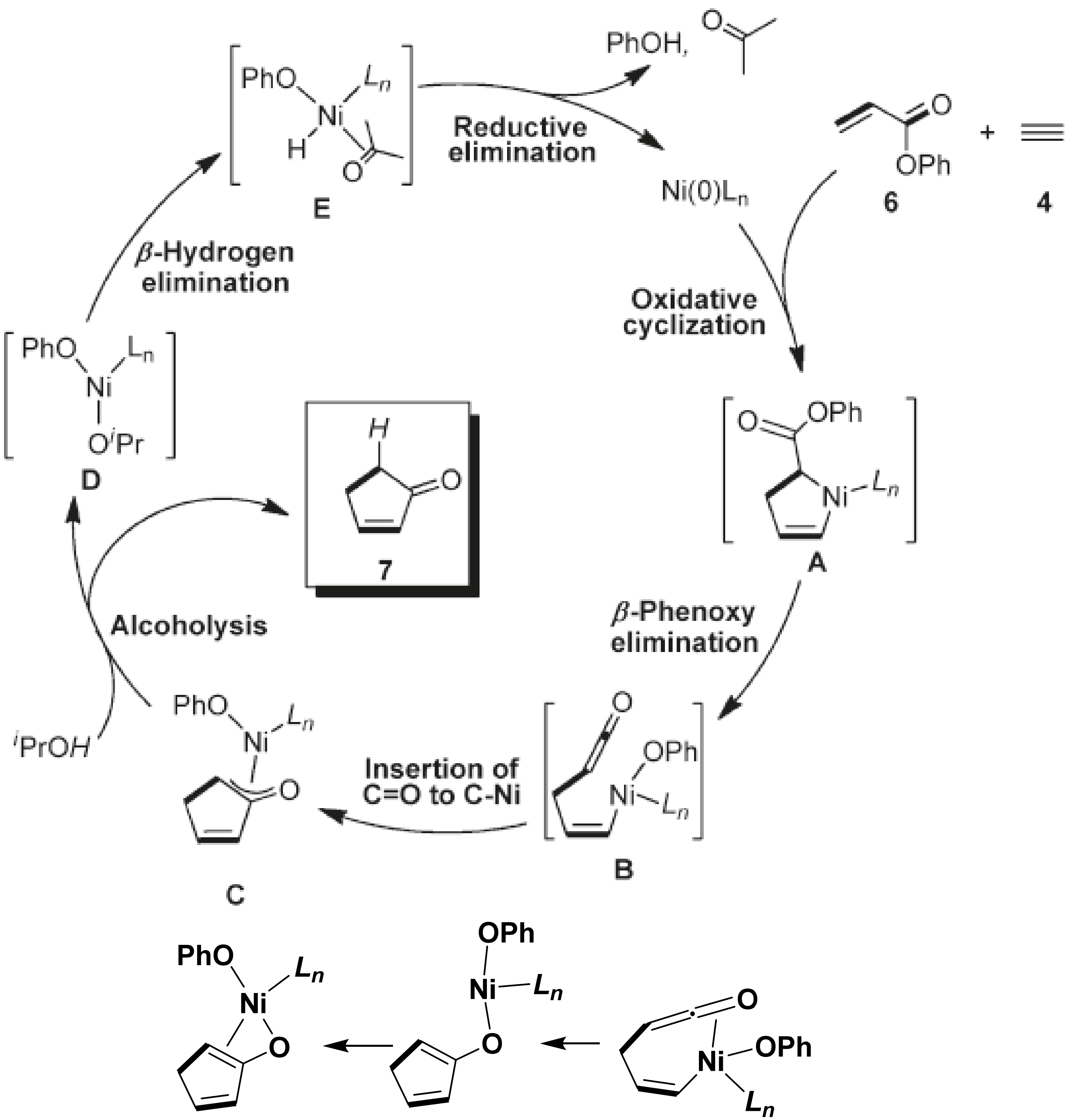
Masato Ohashi,*[†,‡] Tomoaki Taniguchi,[†] and Sensuke Ogoshi*[†]

†Department of Applied Chemistry, Faculty of Engineering, and ‡Center for Atomic and Molecular Technologies, Osaka University, Suita, Osaka 565-0871, Japan



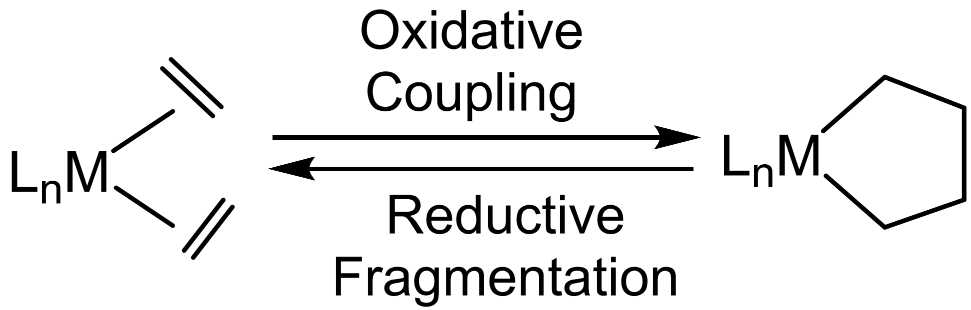
Oxygen-containing organic compounds, such as ethers, carboxylates, and carbamates, have received increasing attention because of their newly discovered applications as electrophiles in cross-coupling reactions via transition metal-catalyzed CO bond activation.
However, no cycloaddition reaction involving their CO bond activation has been demonstrated thus far. The present study developed a Ni(0)-catalyzed unique [3+2] cycloaddition reaction of α,β-unsaturated phenyl esters with alkynes in iPrOH to yield cyclopentenone derivatives. Transition

J. Am. Chem. Soc., **2011**, *133*, 14900



PhOH,
Reductive elimination
Ni(0)L_n
6
4
Oxidative cyclization
A
β-Phenoxy elimination
B
Insertion of C=O to C-Ni
C
^{i}PrOH
Alcoholysis
D
β-Hydrogen elimination
E
7

Reductive fragmentation (还原碎片化)

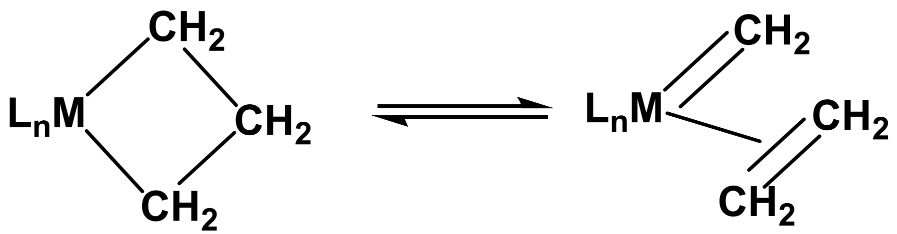


The reductive fragmentation is much rarer. It cleaves a relatively unactivated C–C bond to give back the two unsaturated ligands.

From the point of view of the metal, these are cyclic OA and RE reactions; the new aspect is the remote C–C bond formation or cleavage.

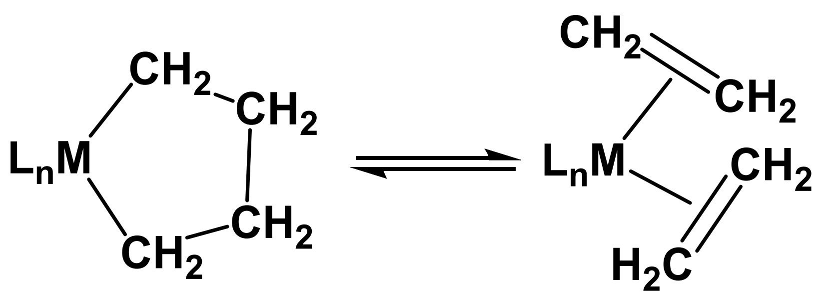
Recall the special properties of some M-R complexes

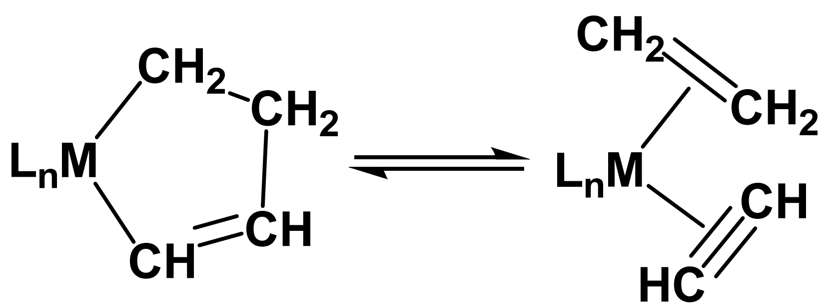
Small metallocylcles show interesting rearrangement.



Precusors to carbene

reductive fragmentation





Larger metallacylcles show normal properties of alkyls, e.g.